Looking Ahead? The Role of Future Educational Opportunities on Pre-College Human Capital

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Abstract

How do future schooling opportunities shape early educational decisions? In this paper, I exploit a large expansion in the supply of free higher education institutions (HEIs) in Brazil to analyze whether tertiary education opportunities can shift demand for secondary schooling in low- and middle-income countries (LMICs). To understand when higher education expansion can increase pre-tertiary attainment, I develop a simple framework that highlights the channels of impact and necessary conditions for downstream effects to occur. I construct a novel dataset on HEI rollout and combine it with rich administrative data to estimate the effect of receiving an HEI on local public-school students' secondary-school persistence using a staggered difference-in-differences design. Following the opening of a new campus, dropout rates among high school students fall by 1 percentage point (a 5-6% decrease from the baseline mean). As a result, students in treated municipalities are also 2% more likely to graduate from high school, corresponding to over 20,000 additional high school graduates across the first three exposed cohorts. I show that these effects are driven by increased access to higher education rather than by local labor market changes, with local students' enrollment in federal HEIs increasing by 140% after the policy. As the framework predicts, the college premium plays a key role in generating these downstream effects: changes in high school graduation are positive only in municipalities with a high pre-policy college premium, and largest where the college premium is high and the high school premium is low. Expanding access to higher education in contexts where returns to education are highly convex can be a powerful tool for improving early educational attainment, especially in LMIC settings where stand-alone returns to high school are not enough to keep students in school.

Keywords: Returns to education, higher education, college premium, educational attainment

JEL Classification: O12, O15, O54, I25, I26

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1 Introduction

Barriers to education in low- and middle-income countries (LMICs) have declined substantially in recent decades, yet a significant share of age-eligible students remains out of school. By 2023, over half of all countries included free primary and secondary education in their legal frameworks (UNESCO, 2025). In Brazil, the context of this paper, primary and secondary education has been free for over 40 years and became compulsory in 2009. Yet in 2024, over 25% of 24-to-35-year-olds in Brazil did not complete high school (OECD, 2026) and over 1 in 10 high school students dropped out before graduating.

Low returns to education do not obviously account for Brazil's weak demand for schooling. In the 2000s, estimated Mincerian returns to one extra year of schooling were nearly 10% (Firpo and Portella, 2024), yet adults attained fewer than six years of schooling on average. Returns to education are, however, also highly convex and much larger at later stages that most students never reach: in LMICs, tertiary returns are two to three times the secondary premium (Gethin, 2025) and, in Brazil, college graduates earned 140% more than high school graduates in 2010. Expanding access to these otherwise unreachable returns at the top of the schooling distribution could pull constrained students up the educational ladder. In this paper, I ask: what happens when higher education becomes more accessible to all?

I explore an ambitious federal program in Brazil between 2003 and 2014 that more than doubled the supply of public and free four-year colleges and two-year trade schools. I evaluate whether the arrival of a new higher education institution (HEI) in a municipality affects local public students' pre-tertiary persistence, measured by the likelihood of starting and graduating high school, and dropout rates from 9th to 12th grade. In doing so, I assess whether investments at the top of the schooling distribution can raise overall demand for education[1].

It is not clear what might happen to early educational attainment when college supply increases. On one hand, lower college attendance costs and a higher likelihood of accessing college returns might lead students to complete basic education at higher rates. On the other hand, changes in the local labor market that increase demand for unskilled labor can increase the opportunity costs of staying in school. To understand how and when higher education expansion can influence pre-tertiary educational decisions, I develop a simple framework that reconciles mixed evidence stemming from different contexts. I identify the necessary conditions that must be in place for

[1]The policy includes expanding both universities and trade schools, which traditionally offer technical and professional courses, and majors focused on technical skills, although some do offer traditional majors.

students to change their pre-tertiary schooling decisions after new higher education opportunities are introduced. I also identify two channels through which a new HEI might shift the share of students completing high school: a direct *educational channel*, by increasing the probability of entering tertiary education and decreasing tertiary education attendance costs, and an indirect *labor market channel*, by changing local wage levels. My framework highlights the college premium's key role in driving high school graduation by increasing expected returns to high school, especially where the standalone high school premium is modest. As a result, the same higher education expansion can raise high school completion in some places and have little effect in others, depending on the relative size of local college and high school premiums.

Empirically, I estimate the impact of higher education expansion on pre-tertiary students' outcomes using a staggered difference-in-differences approach proposed by Callaway and Sant'Anna (2021), comparing public school students in municipalities that received a new HEI with those in comparable municipalities that never received one. I build a novel dataset of over 400 HEI openings using information from institutional documents, HEIs' websites, and legal ordinances, combined with rich administrative data from the Ministry of Education covering the universe of K-12 enrollment and of individuals who take the national college-entry exam.

There are three key challenges for my identification strategy. First, treated municipalities were not randomly selected by the federal government and could be on a different trend than control municipalities. To address this selection problem, I control for a number of municipality-level characteristics observed by the government when choosing receiving municipalities as stated in institutional documents, and relevant characteristics documented in the literature, such as political alignment (Arroyo and Petterini, 2020). Event-study estimates show no evidence of differential pre-trends. Second, the government may have targeted municipalities based on their pre-policy educational outcomes. Pre-treatment outcome levels either do not predict treatment or are somewhat worse in treated municipalities. Since my design compares changes rather than levels, these baseline differences do not bias my estimates on their own. Third, the large influx of students and staff resulting from this policy can alter the composition of primary and secondary schools, potentially driving my results. I investigate whether the shares of white students, students who attended only public schools, and students whose mothers completed at least high school are changing as a result of this policy. I find no evidence that changes in composition could explain the observed results. Finally, my results are also robust to alternative definitions of the control group and of the high school graduation outcome, to different sets of control variables (including none), and to

a permutation test, which shows that the estimated effects are unlikely to arise by chance.

I find that public school students in municipalities that received an HEI as part of this program increased persistence at all three grades of high school by 5-6% relative to the baseline mean, with no effect on dropout rates at 9th grade, the last grade of primary education. This increase in persistence translates into higher high school completion: students were 2% more likely to eventually graduate from high school, but not more likely to graduate on time. This amounts to over 20,000 additional high school graduates in the first three cohorts exposed to a new HEI. Students were also 20% more likely to take a national college-entry exam, with no decrease in scores. I find no significant impacts on in-school performance, measured either at 9th grade or at the end of high school.

This policy provides a unique setting where different municipalities received different types of HEIs. Comparing policy effects by type of HEI received allows me to test whether different types of post-secondary education affect pre-tertiary students' decisions in different ways. I find suggestive evidence that, while both universities and trade schools increase high school completion, only universities appear to change the decision to enroll in high school in the first place. This suggests that the two types of institutions move different marginal students. Trade school opportunities seem more salient to students who would have transitioned to high school but dropped out in the absence of the policy, but now choose to stay the course and graduate. Universities might change a student's decision to start high school in the first place.

Throughout, all of my analysis focuses on public school students, who are more likely to come from low-income households. However, within public schools, non-white students are especially disadvantaged on all metrics of pre-tertiary persistence and performance. I further analyze heterogeneity by race to identify who is most likely to benefit from the policy. I find that only non-white students respond to new higher education opportunities by changing their decision to start high school and, conditional on doing so, experience a 4% increase in the likelihood of graduating high school, nearly four times the effect observed for white students (although only marginally statistically different). The observed effect on the likelihood of starting high school and completing high school closes 1/3 of the pre-policy gap between white and non-white students in these two metrics.

The average 2% increase in high school graduation following this higher education policy represents one-eighth to one-sixth of the effects found in quasi-experimental studies assessing the impact of being admitted to secondary school or receiving a secondary school voucher on graduation (Ozier, 2018; Angrist et al., 2006). Although the effect is seemingly small in magnitude, this policy did

not change the supply of high school opportunities or directly target high school completion. I use the theoretical framework's predictions around high school graduation to conduct empirical exercises that provide suggestive evidence on the key channels of impact that might be mediating the observed effect.

While the framework identifies two theoretical channels of impact, I find that the observed high school graduation effects are likely driven by changes in direct educational opportunities rather than indirect changes in the local labor market. Local public school students experience a nearly 1-percentage-point increase in enrollment following the opening of an HEI, representing a 140% increase relative to the pre-policy mean. On the other hand, I find no significant changes in formal-sector wages that could account for the observed positive effects on school completion. As predicted by the framework, changes in high school completion following an expansion in higher education opportunities should be most prevalent in contexts where the college premium is high, and the high school premium is modest: that is, where a new HEI offers the largest expected gain to staying in school. I empirically assess this claim and find that high school graduation effects are positive only in municipalities with a high college premium pre-policy, and largest in municipalities with a high college premium and low high school premium.

Last, to assess whether higher education opportunities might be a sustainable channel to increase returns to high school, I examine how college and high school premiums in the formal labor market change over the 6 years following the opening of an HEI. I find that college premiums remain stable over time, but high school premiums decrease sharply following the opening of an HEI. This drop in high school premiums is driven by an increase in primary-educated wages rather than a decrease in high school-graduate wages, which might be a direct result of a decrease in the supply of unskilled labor due to changes in educational attainment stemming from this policy, a result similar to that found by Khanna (2023) for primary education expansion in India. Provided that the college premium remains sizeable (a reasonable assumption given the limited supply of higher education in Brazil, where the share of adults who completed college in 2000 was 7%), higher education opportunities should continue to shape pre-tertiary schooling decisions in the long run.

This paper contributes to the broader literature on the effects of college expansion or proximity on educational attainment, which primarily focuses on the direct impact through college enrollment and completion, and local economic impacts (Currie and Moretti, 2003; Lochner and Moretti, 2004; Grossman, 2006; Oreopoulos and Petronijevic, 2013; Vu, 2024; Feinmann and Hsu Rocha, 2026; Cox, 2024). I add to this literature by providing evidence on the effects of *expected* higher education

attendance on local teenagers' human capital investments, a result especially relevant for low- and middle-income countries where average attainment falls well below high school completion.

I directly build on and contribute to the literature examining the effects of subsequent educational opportunities on earlier educational attainment. While a strand of this literature examines the effects of affirmative action on secondary students' outcomes (Akhtari et al., 2023; Khanna, 2020), my paper more closely aligns with a growing literature on the impact of expanding the supply of higher levels of education on students' attainment at lower levels (Bedard, 2001; Jagnani and Khanna, 2020; Katzkowicz et al., 2023; Sandholtz, 2024; Rands and Barsanetti, 2025; Erbabian, 2026). While positive impacts from higher education expansion on pre-tertiary schooling outcomes are observed in India (Jagnani and Khanna, 2020) and Kenya (Erbabian, 2026), negative impacts are documented in the United States (Bedard, 2001) and Chile (Erbabian, 2026).

I build on this literature in a few ways. First, I develop a framework that reconciles this mixed evidence across different contexts. In my framework, I formally show how the pre-existing college and high-school premiums, together with baseline high school completion rates, determine when higher education opportunities positively affect pre-tertiary attainment, and when negative downstream effects may arise. I further illustrate the possible channels of impact (via higher education opportunities versus broader labor market changes) and empirically test whether the framework's predictions hold in my setting, and which mechanisms drive the observed effects. Second, I provide novel evidence on the types of higher education institutions that matter for teenagers' schooling outcomes. In particular, I show that not only traditional universities but also trade schools can shift local students' high school decisions.

The remainder of this paper is organized as follows. Section 2 details the HEI expansion policy. Section 3 presents a theoretical framework to explain how the expansion of tertiary education can affect students' educational decisions at lower levels. Section 4 describes the data and provides more detail on the placement of HEIs. Section 5 describes the method used in this paper. Section 6 reports the results for all pre-tertiary outcomes. Section 7 provides empirical evidence on the mechanisms through which the policy operates, while Section 8 examines the policy's long-run impact on local wages. Section 9 presents robustness analyses, and Section 10 concludes with a discussion of the results.

2 Background

2.1 Higher education expansion program in Brazil

Tertiary education in Brazil comprises public and private institutions. The former is primarily provided by the federal government through Federal Universities and Trade Schools.[2] These institutions are free to students, and admissions are primarily based on a national exam (ENEM) test score that students completing high school can take once a year.[3] After obtaining their college-entry exam test scores, students apply to federal universities through the Unified Selection System, SISU, a centralized platform that allows students to choose the major and HEI combination they wish to pursue. Private and state universities have their own admission process, although some also use ENEM test scores.

In 2003, the federal government of Brazil launched a large higher education expansion program, with the main goal of reducing the concentration of higher education institutions in higher-income regions and capitals, and using higher education to promote regional economic development, aiming to increase municipalities' income and reduce regional inequalities. The government selected municipalities based on geographical and economic criteria, such as the municipality's Human Development Index (HDI), population size, local economic activities, and pre-existing supply of higher education institutions (Barbosa et al., 2020; Ministério da Educação [MEC], 2011). The policy included the expansion of new public and free universities and trade schools, as well as new campuses of already existing HEIs, and was divided into three phases.

Figure 1 presents the distribution of federal university campuses before and after the expansion program and illustrates the policy's goal of decreasing the concentration of higher education institutions in capitals, metropolitan areas, and more developed regions of the country.

The first expansion phase, called Expansion I, occurred between 2003 and 2007, and its main goal was to expand higher education institutions into municipalities in the inner part of the country, thus earning the name "interiorization phase". During this period, 98 institutions were created. The second expansion phase, called REUNI, occurred between 2008 and 2012 and combined the continuation of the expansion started in 2003 with the restructuring of existing universities, along with expanded enrollment capacity and infrastructure improvements, including new buildings and

[2] State governments can also create and fund state universities, which are also free to students.

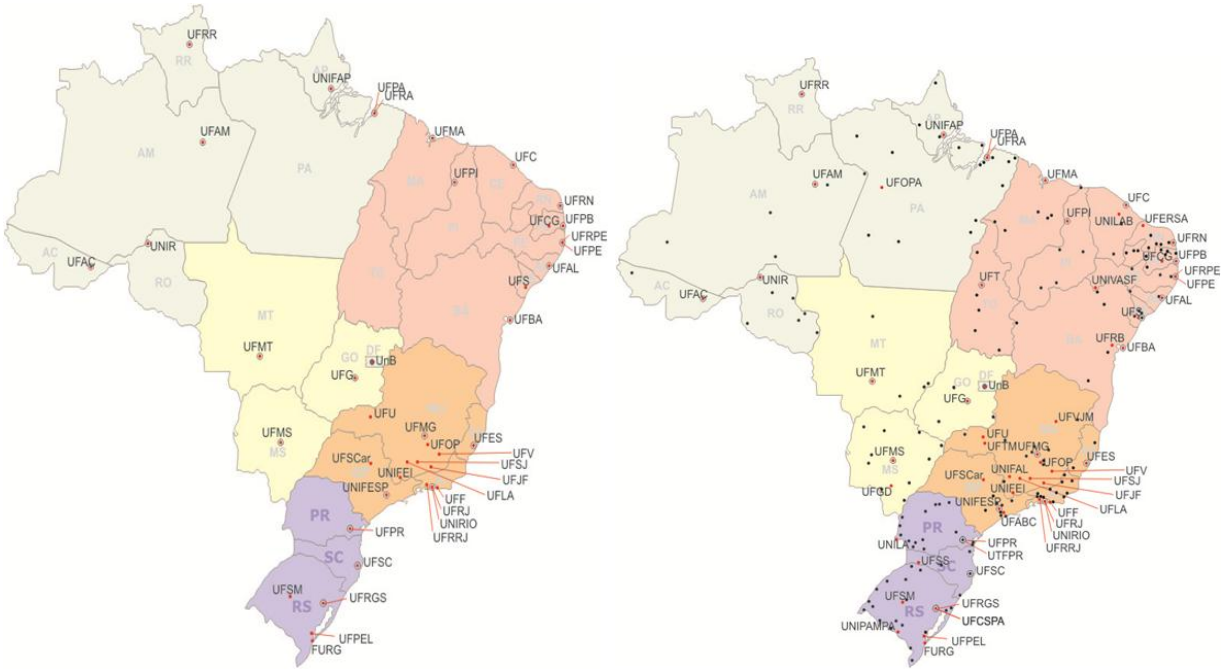
[3] The national unified exam, ENEM, started being used in college admissions in 2010. Beforehand, students took individual exams for each university they would apply to. Trade schools have alternative admission routes, but also use ENEM as part of their admissions process.

classrooms. In this second phase, 247 new institutions were created.

The third phase of the expansion occurred between 2012 and 2014, although institutions continued to open up until 2017. It focused on coupling higher education expansion with initiatives to develop underserved regions and on launching an array of public programs, such as policies to expand medical school enrollment and to include people with disabilities in higher education. All these additional policies targeted college students enrolled in these newly opened institutions. During this phase, 239 HEIs were created.

Throughout these three phases, 554 municipalities (nearly 10% of all Brazilian municipalities) received at least one institution between 2003 and 2017. Of those, 252 municipalities received their first higher education institution.

Figure 1: Expansion of Federal Universities from 2002 (left) to 2014 (right)



Note: This figure comes from the Brazilian Ministry of Education website. Distribution of federal universities in Brazil in 2002 (left), with 45 institutions in 141 municipalities, with a total of 151 campuses, and in 2014 (right), with 63 institutions in 275 municipalities, with a total of 321 campuses. Colors represent the different regions of Brazil. This map does not include Trade Schools. Before the policy, there were 140 federal trade school campuses in Brazil.

2.2 Federal trade schools

Trade schools account for approximately two-thirds of the HEIs opened under the federal expansion program. These institutions differ from universities in relevant ways that could generate differential effects, depending on the type of HEI a municipality receives. Trade schools are part

of the professional and technological education programs in Brazil and were first conceived in the early 1900s as schools to provide lower-income individuals with technical skills, such as carpentry. Between 1930 and 1970, the federal government expanded the supply of technical education to meet the growing demand for middle-skilled labor arising from industrialization, particularly for employment in the automobile and petrochemical industries. At this time, trade schools started providing a high school-equivalent education to attract young individuals (Faveri et al., 2018).

In the 1980s and 1990s, trade schools began offering training in computational skills and information technology (IT) to keep pace with changes in labor demand. Up until 2000, they remained focused on high school-equivalent education and short-term technical courses targeting hard skills. They were then incorporated into the Brazilian Higher Education system, although institutes still offer technical training that does not require a high school diploma. Since then, trade schools have offered three main types of courses: integrated technical courses, in which students are enrolled in high school and a technical course simultaneously, and which last 3 to 4 years; subsequent technical courses, with a minimum duration of 2 years, in which students need to have a high school diploma to enroll; and undergraduate majors, which are similar to traditional universities[4]. Trade schools can also offer short-term courses, such as elderly caregiving and basic English, as well as graduate education, such as a Master's in Materials Engineering.

3 Why and when do higher education opportunities matter? A theoretical framework

Given that the provision of higher education does not directly affect the availability or costs related to secondary schooling, it is important to understand when and why an increased supply of subsequent educational opportunities can change earlier educational decisions. In this section, I develop a simple model to explain how students decide whether to complete high school and how the availability of higher education factors into this decision. This model has two goals: first, to identify the channels through which the availability of higher education influences the decision to graduate from high school, and the specific parameters that may be affected by an increased supply of local HEIs. Second, to identify the necessary conditions that need to be in place for trickle-down educational effects to occur, which will be key to reconciling mixed evidence stemming from different

[4]Examples of integrated technical courses are IT and construction. Examples of subsequent technical courses are business administration, workplace safety, and nursing technician. Examples of undergraduate majors are energy engineering, architecture, and math.

contexts, such as Jagnani and Khanna (2020) and Erbabian (2026) for India and Kenya, and Bedard (2001) and Erbabian (2026) for the United States and Chile, and generate predictions of what can be expected in the Brazilian setting.

3.1 A student's decision: baseline model

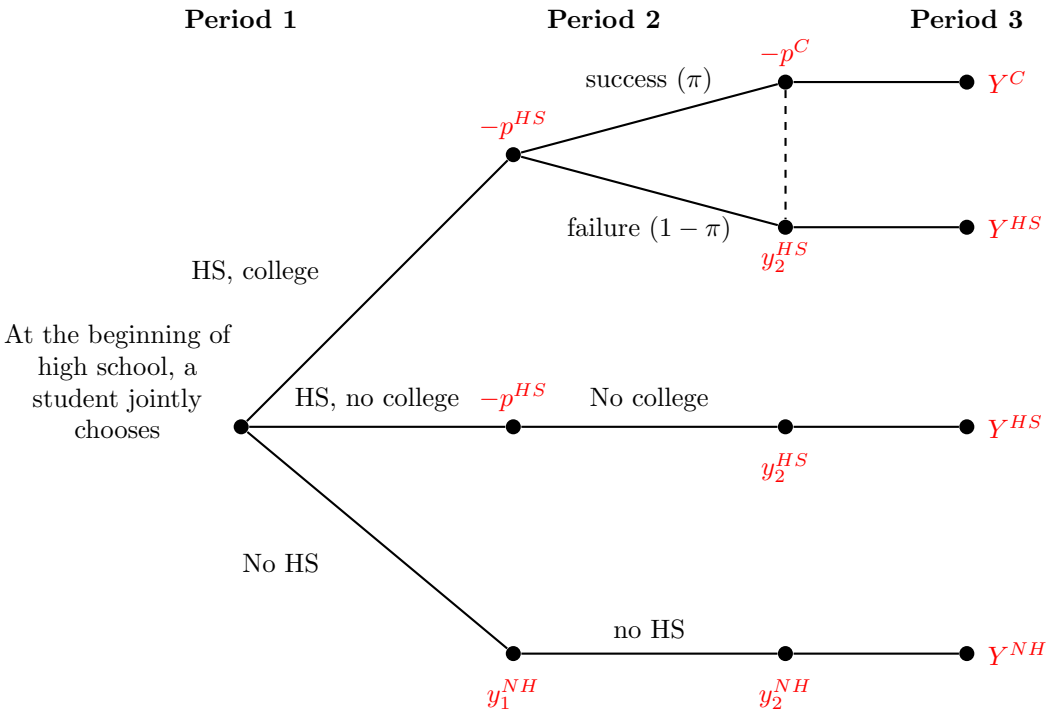
Figure 2 illustrates the decision of a forward-looking student just starting 10th grade, the first year of high school in the Brazilian system. At that point, a student must simultaneously decide whether to continue in and graduate from high school and whether to *attempt to* graduate from higher education. Three possible paths are available to a student. If a student decides to drop out of high school, they automatically choose not to pursue higher education, follow the (no high school, no college) path, and immediately join the labor force in period 1. On the other hand, if a student graduates from high school, they remain in high school in period 1, and then follow one of two paths depending on their higher education decision. If choosing not to attend college, they follow the (high school, no college) path and join the labor force in period 2, immediately after completing secondary education. If choosing the (high school, college) path, they will successfully enroll in and graduate from higher education in period 2 with probability π_c, and fail and join the labor force with a high school degree $(1-\pi_c)$ – similarly to those who chose the (high school, no college) path[5]. This uncertainty is key to modeling the Brazilian setting, where public higher education spots are limited, acceptance rates are below 10%, and student loans to attend private higher education were uncommon for most of the policy period. Lower-income public school students, the focus of this paper, therefore face significant constraints in attending higher education and may not attend even after choosing to try.

In this setting, both high school and college attendance incur transportation/relocation and material costs, and a student must pay p^{HS} and p^{C} if continuing in high school in period 1 and higher education in period 2, respectively. For simplicity, I abstract from tuition-related costs. Since public high school is readily available and HEIs opening as part of the Brazilian policy are free, this is a reasonable abstraction[6]. When a student enters the labor force, they start earning a constant wage specific to their final educational level: y^{NH} for high school dropouts, y^{HS} for high

[5]In this framework, I will abstract away selection issues that would lead high school-level earnings to be different for a student who chose high school and no college, and a student who chose high school, college but failed to progress to higher education.

[6]Private higher education is also a relevant share of tertiary institutions in Brazil. For simplicity and due to data availability, I do not directly model private HEI enrollment as a student outcome. For my sample, which consists only of public school students coming from lower-income households, and for most of this time period, when loans to attend private colleges were limited, this is an acceptable simplification.

Figure 2: Decision tree with joint choice of high school and college attendance



school graduates, and y^C for higher-education graduates.

In a setting where higher education is not available or where distance and moving costs are so high for the median student that college is not in their choice set, students will choose only between (high school, no college) and (no high school, no college), or simply whether to graduate high school. In this case, if the high school premium is high enough to justify the foregone earnings and attendance costs in period 1, students will choose to stay in school rather than drop out and join the labor force. Now consider a world where higher education is available. In addition to the high school decision, a high school freshman will also choose whether to pursue higher education after graduation. In this scenario, choosing to graduate high school opens an additional path, (high school, college), which might lead to greater lifetime earnings.

The decision to attend high school varies greatly depending on which world a student is in. Assuming students have certainty about wages by education level, when higher education is not an option, the expected earnings from graduating high school are simply the high school wage, and the schooling decision becomes a simple comparison of lifetime earnings with and without graduating high school. On the other hand, with the added possibility of pursuing higher education (and thus accessing higher-education wages), students now face uncertainty about their future earnings

when choosing the path corresponding to (high school, college). Or: $\mathbb{E}[y^{HS}_{hs,nocol}] = y^{HS}$, while $\mathbb{E}[y^{HS}_{hs,col}] = (1-\pi_c)y^{HS} + \pi_c y^{C}$. Simply put, this additional educational opportunity represents a potential increase in the expected returns to graduating high school, provided that wage levels vary considerably by educational level in the local labor market.

I formalize the decision-making process illustrated in Figure 2 as a simple three-period problem described below. The student chooses their educational path from $(s,d) \in \{(0,0),(1,0),(1,1)\}$, as defined above. Because graduating from college requires first graduating from high school, the pair $(0,1)$ is not feasible. The student maximizes the expected utility:

$$\max_{(s,d)\in\{(0,0),(1,0),(1,1)\}} U(C_1) + \beta_2\,\mathbb{E}\big[U(C_2)\big] + \beta_3\,\mathbb{E}\big[U(C_3)\big], \tag{1}$$

where $u = ln(c)$ is a standard strictly increasing and strictly concave utility function and $\beta_2 = \beta^3$ and $\beta_3 \equiv \sum_{t=8}^{N}\beta^{t-1}$. β is the student's discount factor. The consumption streams for each period are:

Period 1. Period 1 lasts 3 years, the duration of high school in Brazil, grades 10 to 12:

$$C_1 = \bar{k}_1 + (1-s)\,y_1^{NH} - s\,p^H,$$

where $\bar{k}$ is the level of household assets, which includes the earnings of other family members, $\bar{k}_1 = 3\bar{k}$ and $y_1^{NH} = 3y^{NH}$.

Period 2. Period 2 lasts 4 years, the typical duration of tertiary education in Brazil:

$$C_2 = \begin{cases} \bar{k}_2 + y_2^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted with probability } (1-\pi_c), \\ \bar{k}_2 - p^c & \text{if } (s,d) = (1,1) \text{ and admitted with probability } \pi_c. \end{cases}$$

where $\bar{k}_2 = 4\bar{k}$, $y_2^{NH} = 4y^{NH}$ and $y_2^{HS} = 4y^{HS}$.

Period 3. Period 3 summarizes the remainder of an individual's working life as the present value of the per-year consumption C_3. After period 2, an individual's education level is fixed,

and consumption will be constant for the following N periods during which the individual remains employed.

$$C_3 = \begin{cases} \bar{k} + y^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted}, \\ \bar{k} + y^{C} & \text{if } (s,d) = (1,1) \text{ and admitted}, \end{cases}$$

For simplicity, I assume $\bar{k}$ to be constant over time. Similarly, earnings by level of education are constant and known to students at the time of the schooling decision, and students' final educational attainment cannot change after this initial decision and the realization of the college draw. For paths $(0,0)$ and $(1,0)$, there is no uncertainty given that the student will not apply to college, and the only source of uncertainty is the college admission probability, $\pi_c \in (0,1)$.

Costs and earnings incurred by the students, as well as the timing for when they will join the labor force, are as described above. For simplicity, I will assume that costs are drawn independently and uniformly: $p^H \sim U[0, \bar{P}^H]$ and $p^c \sim U[0, \bar{P}^c]$. I further assume that $\bar{P}^H < \bar{k}_1$ and $\bar{P}^c < \bar{k}_2$ so that period-1 and period-2 consumption are strictly positive on all paths. A student will decide whether to follow $(0,0)$, $(1,0)$, or $(1,1)$ by comparing the expected lifetime utility they will receive from each path. Using $U(c) = \ln(c)$, the lifetime utilities of the three paths are:

$$V(0,0) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}), \tag{2}$$

$$V(1,0) = \ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}), \tag{3}$$

$$\begin{aligned} V(1,1) = \ln(\bar{k}_1 - p^H) &+ \beta_2 \big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] \\ &+ \beta_3 \big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^{C})\big]. \end{aligned} \tag{4}$$

By setting $V(1,0) = V(0,0)$, $V(1,1) = V(0,0)$ and $V(1,1) = V(1,0)$, I derive thresholds for when a student will prefer $(1,0)$ over $(0,0)$ (threshold $\bar{p}^H_{10}$), $(1,1)$ over $(0,0)$ (threshold $\bar{p}^c_{11|00}$) and $(1,1)$ over $(1,0)$ (threshold $\bar{p}^c_{11|10}$), respectively. The detailed derivation of these thresholds is in the

Appendix. Combining the three thresholds, the student's decision rule is:

$$(s,d)^* = \begin{cases} (1,1) & \text{if } \left[p^H \leq \bar{p}^H_{10} \text{ and } p^c \leq \bar{p}^c_{11|10}\right] \text{ or } \left[p^H > \bar{p}^H_{10} \text{ and } p^c \leq \bar{p}^c_{11|00}\right], \\ (1,0) & \text{if } p^H \leq \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|10}, \\ (0,0) & \text{if } p^H > \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|00}. \end{cases} \tag{5}$$

where:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}. \tag{6}$$

$$\bar{p}^c_{11|10} = \bar{k}_2 - (\bar{k}_2 + y_2^{HS}) \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^{C}}\right)^{\beta_3/\beta_2}. \tag{7}$$

$$\bar{p}^c_{11|00} = \bar{k}_2 - \left[\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right)^{1/\beta_2} \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^{C})^{\pi_c}}\right)^{\beta_3/\beta_2}\right]^{1/\pi_c}. \tag{8}$$

Each threshold marks the point where a student is indifferent between two options: at $p^H = \bar{p}^H_{10}$, a student is indifferent between $(0,0)$ and $(1,0)$; at $p^c = \bar{p}^c_{11|10}$, a student is indifferent between $(1,1)$ and $(1,0)$; and at $p^c = \bar{p}^c_{11|00}$, a student is indifferent between $(1,1)$ and $(0,0)$. In the absence of higher education, the relevant threshold would only be $\bar{p}^H_{10}$, which mediates the choice between (no high school, no college) and (high school, no college) and simply compares the value of staying in high school versus dropping out. With the added higher education opportunity, it is now possible for students who would drop out of high school via $\bar{p}^H_{10}$ to choose the (high school, college) path through the $\bar{p}^c_{11|00}$ threshold: for those with prohibitive p^H to choose high school alone, high school now becomes attainable when $p^c \leq \bar{p}^c_{11|00}$.

Let S^* denote the aggregate share of students who graduate high school. A student graduates high school either by choosing $(1,0)$ (so $p^H \leq \bar{p}^H_{10}$) *or* $(1,1)$ over $(0,0)$ (so $p^H > \bar{p}^H_{10}$ and $p^c \leq \bar{p}^c_{11|00}$). Hence:

$$S^* = \Pr(p^H \leq \bar{p}^H_{10}) + \Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \leq \bar{p}^c_{11|00}\big). \tag{9}$$

Applying the law of total probability and independence of p^H and p^c:

$$S^* = \Pr(p^H \leq \bar{p}^H_{10}) + \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c \leq \bar{p}^c_{11|00}(p^H)\big) \cdot f(p^H)\, dp^H,$$

where $f(p^H) = 1/\bar{P}^H$. Under $p^c \sim U[0, \bar{P}^c]$, $\Pr(p^c \leq \bar{p}^c_{11|00}(p^H)) = \bar{p}^c_{11|00}(p^H)/\bar{P}^c$. Therefore:

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{10}$$

The first term represents the share that finds high school worthwhile on its own, while the second is the share that would not graduate high school alone but is induced to complete the full college path when college costs are low enough. Note that S^* does *not* depend on $\bar{p}^c_{11|10}$, the threshold pertaining to students who move from $(1,0)$ to $(1,1)$.

Given the composition of S^* described above, changes in the share of students completing high school can stem from two distinct transitions. The first transition moves students from (no high school, no college) to (high school, no college). The threshold which mediates this transition, $\bar{p}^H_{10}$, depends only on household assets and income, $\bar{k}$, and primary-educated and high school-graduate wage levels, y^{NH} and y^{HS}. Importantly, it does not depend on parameters related to higher education opportunities that might be directly affected by HEI expansion, such as college attendance costs, p^C, and the likelihood of attending an HEI, π_c. Thus, movements from $(0,0)$ to $(1,0)$ will result solely from changes in the local labor market brought about by these new institutions, and are not a product of a direct increase of future educational opportunities. As Jagnani and Khanna (2020) and Lallemant and Dillon (2026) have documented important infrastructure and labor market changes in places receiving new HEIs in India and Uganda, respectively, it is worth examining the role of this more indirect channel in shaping pre-tertiary schooling decisions.

The second, and more direct transition moves students from the (no high school, no college) path directly into (high school, college), and operates through the $\bar{p}^c_{11|00}$ threshold. Equation 8 shows that $\bar{p}^c_{11|00}$ depends not only on earnings by education levels and household wealth, but also on college attendance costs (via the support $\bar{P}^c$) and the probability of enrolling in higher education, π_c. These movements, although also affected by changes in the local labor market, are more closely related to the policy's direct effect. Here, the college attendance decision directly alters the high school decision: students who would not have graduated high school alone (i.e., would not choose (high school, no college) path) are now driven to attend high school *because* of the possibility of

attending college.

If we assume that, at least in the short run, local wage levels do not change following the opening of an HEI, all movements in S^* would stem from students transitioning from $(0,0)$ to $(1,1)$ and operate through an increase in educational opportunities, here represented by parameters p^C and π_c. While I will empirically assess whether this is true, for completeness, I will allow wages to change with the policy and examine the implications of these changes on schooling decisions.

3.2 Necessary conditions: When can college supply impact pre-tertiary educational attainment?

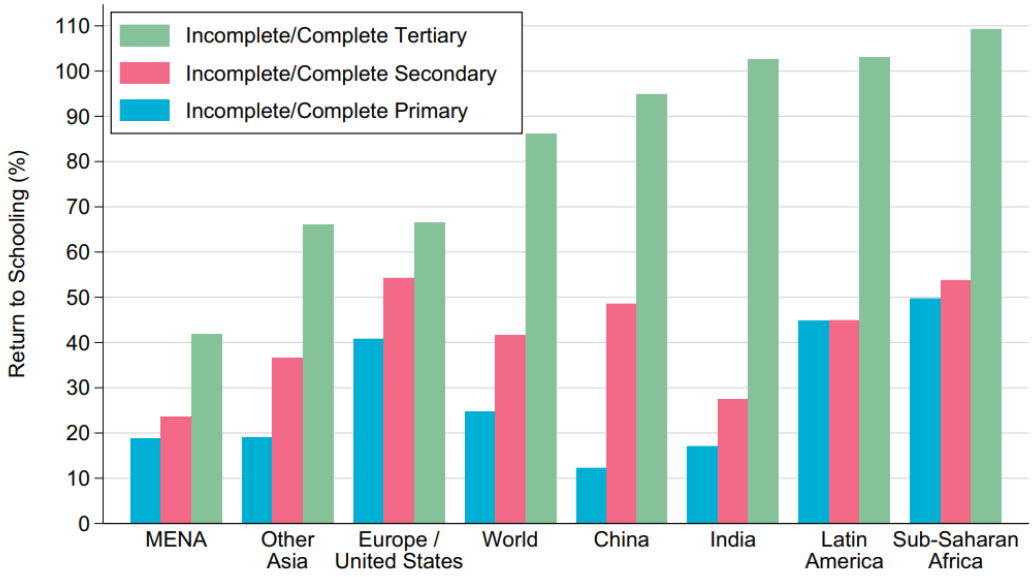
Three main conditions need to be in place for the higher education expansion to increase secondary school completion rates. First, a necessary condition for either transition to be possible is that there is a mass of students who would choose not to graduate from high school in the absence of higher education. That is, at least some students were choosing the path (no high school, no college) prior to the opening of an HEI. Since students will choose this path only when $p^H > \bar{p}^H_{10}$, this translates into a condition on the threshold $\bar{p}^H_{10}$: $\bar{p}^H_{10} < \bar{P}^H$, which simply implies that it needs to be possible for some students to be above the threshold. As shown in the Appendix, this condition fails whenever high school wages tend to infinity. In practice, this implies that in places where the high school premium is very high, students will always choose to graduate from high school, and new HEIs will have no one to move into either of the two high school paths. This necessary condition then imposes an upper bound on the high school wage premium, and establishes the first prediction of the framework: places with higher baseline high school graduation should see smaller responses in high school completion following the opening of an HEI.

The second condition pertains only to movements from (no high school, no college) to (high school, no college) – the indirect transition driven only by changes in the local labor market. For students to be moved to (high school, no college), the threshold governing that transition, $\bar{p}^H_{10}$, needs to be positive, so that $p^H < \bar{p}^H_{10}$ for at least some students, given that $p^H > 0$. A necessary condition for $\bar{p}^H_{10} > 0$, as shown in the Appendix, is that high school wages must be larger than primary-level wages. This condition allows graduating high school alone to be preferred over dropping out, since students on this margin will only be comparing high school wages and primary-educated wages.

The third condition pertains only to students choosing to graduate high school by moving directly into the (high school, college) path – the direct transition driven by educational opportunities. Similar to the condition above, for students to transition from (no high school, no college) to (high

school, college), it must be true that $p^c < \bar{p}^c_{11|00}$ for at least some students. Since $p^c > 0$, this implies that $\bar{p}^c_{11|00}$ must be strictly positive. In the appendix, I show that this condition implies that higher education earnings need to be not only larger than high school earnings, but also large enough, above a certain threshold, $\bar{y}^C$. Together with the first stated condition, this condition establishes the key prediction of the framework: an increase in college access should only drive students to graduate high school in places where the college premium is high, and high school earnings are modest relative to primary education returns.

Figure 3: Returns by region - Gethin (2025)



Note: This figure comes from Gethin (2025), published in the Quarterly Journal of Economics.

Figure 3 shows substantial variability in returns to schooling by education level across regions. In places like India, Latin America, and Sub-Saharan Africa, tertiary returns greatly exceed those for secondary schooling, whereas primary and high school returns are much closer in magnitude. On the other hand, in high-income countries in Europe and the United States, high school returns are significantly higher than those for primary education, and tertiary education gains represent a much smaller jump than those observed in LMICs. According to this simple model and the necessary conditions described above, settings like India, Latin America, and Sub-Saharan Africa are the most suitable ones for an increase in college opportunities to drive students to graduate high school. This is in line with the observed impacts in the literature for Kenya and India, and the opposite evidence in higher-income settings such as the United States and Chile.

3.3 Comparative statics: What changes when a local college opens?

In this section, I analyze how policy-driven changes in parameters might affect the share of students graduating high school, S^*, through both of the transitions described above. The arrival of a new college in a student's municipality might affect multiple aspects of a student's schooling decision, either through direct changes in parameters related to college attendance or indirect labor market changes. In the model, direct impacts occur through changes in the costs of college attendance and the likelihood that a local student will be admitted, while indirect labor market effects materialize through changes in local wage levels. I will show the direction I expect high school graduation rates to move given changes in relevant parameters. Since S^* is monotonic in $\bar{p}^H_{10}$, $\bar{p}^c_{11|00}$, and $\bar{P}^c$, as shown in the Appendix, I can use these thresholds to conduct comparative statics analysis. In later sections, I will empirically assess which parameters the policy effectively changes.

3.3.1 Cost of attending higher education

Changes in the cost of attending higher education p^c only affect movements from $(0,0)$ to $(1,1)$ and, thus, $\bar{p}^c_{11|00}$. By decreasing the distance to the nearest HEI, the policy lowers the upper support of the distribution of p^c, $\bar{P}^c$, shifting students relative to the fixed threshold. Since shifts in attendance costs amount to shifts in the support $\bar{P}^c$, the effect of changing attendance costs on the share of students graduating high school can be described by the following:

$$\frac{\partial S^*}{\partial \bar{P}^c} = -\frac{1}{\bar{P}^H \bar{P}^{c2}} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H < 0. \tag{11}$$

A reduction in college attendance costs shifts more students below $\bar{p}^c_{11|00}$, moving them from (no high school, no college) to (high school, college). This effect is larger when high school attendance costs are lower, as students driven to attend high school because of new higher-education opportunities will have to bear upfront costs in period 1.

3.3.2 Likelihood of attending higher education

Similar to the cost effect, the likelihood of attending higher education, π_c, runs entirely through movements from no high school to the (high school, college) path. Changes in high school comple-

tion given changes in π_c are given by:

$$\frac{\partial S^*}{\partial \pi_c} = \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}_{10}^H}^{\bar{P}^H} \frac{\partial \bar{p}_{11|00}^c(p^H)}{\partial \pi_c} \, dp^H > 0 \quad \text{when } \bar{p}_{11|00}^c > 0. \tag{12}$$

In places where the college premium is sufficiently high (i.e., the necessary condition $\bar{p}_{11|00}^c > 0$ holds), a higher probability of attending higher education unambiguously increases the share of students who graduate from high school. This effect is larger in places and among individuals for whom attendance costs, both at high school and in higher education, are lower. If the college premium is low such that $\bar{p}_{11|00}^c \leq 0$, changes in the probability of acceptance should not change local teenagers' high school attainment.

3.3.3 College-graduate wages

Similarly, changes in college-graduate wages, y^C, operate solely by moving students from (no high school, no college) to (high school, college). As shown in the Appendix, I can differentiate $V(1,1) - V(0,0) = 0$ (the exact point where $p^c = \bar{p}_{11|00}^c(p^H)$)) to sign $\partial \bar{p}_{11|00}^c(p^H)/\partial y^C$.

$$\frac{\partial \big[V(1,1) - V(0,0)\big]}{\partial y^C} = \frac{\beta_3 \pi_c}{\bar{k} + y^C} > 0, \tag{13}$$

so $\partial \bar{p}_{11|00}^c(p^H)/\partial y^C > 0$ for all p^H.

$$\frac{\partial S^*}{\partial y^C} > 0, \tag{14}$$

When college wages increase, students are more likely to complete high school via the (high school, college) path. This effect is increasing on the likelihood of college acceptance, π_c, since students will only be able to access y^C if accepted into higher education. In the limit, if a student's probability (or perceived probability) of college acceptance is zero, then movements in local college-graduate wages should not affect the high school graduation decision.

3.3.4 High-school-graduate wages

Changes in high school level wages, y^{HS}, affect both transitions into (high school, no college) and (high school, college). Below, I separately evaluate how each will change, and the expected net effect on the share of students graduating high school, S^*.

Effect via transitions from (no high school, no college) to (high school, no college). Writing $R \equiv ((\bar{k}_2 + y_2^{NH})/(\bar{k}_2 + y_2^{HS}))^{\beta_2}((\bar{k} + y^{NH})/(\bar{k} + y^{HS}))^{\beta_3}$ and differentiating $\bar{p}_{10}^H = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})R$ using the log-derivative of R:

$$\frac{\partial \bar{p}_{10}^H}{\partial y^{HS}} = (\bar{k}_1 + y_1^{NH})\, R\left[\frac{4\beta_2}{\bar{k}_2 + y_2^{HS}} + \frac{\beta_3}{\bar{k} + y^{HS}}\right] > 0. \tag{15}$$

If new HEIs increase local high school wages, high school on its own becomes more attractive, raising $\bar{p}_{10}^H$ and moving students from $(0,0)$ to $(1,0)$.

Effect via transitions from (no high school, no college) to (high school, college). From (B21) and (B20) in the Appendix:

$$\frac{\partial\big[V(1,1) - V(0,0)\big]}{\partial y^{HS}} = (1 - \pi_c)\left[\frac{4\beta_2}{\bar{k}_2 + y_2^{HS}} + \frac{\beta_3}{\bar{k} + y^{HS}}\right] > 0, \tag{16}$$

so $\partial \bar{p}_{11|00}^c(p^H)/\partial y^{HS} > 0$ for all p^H.

An increase in high school wages increases the expected returns to the (high school, college) path, driving more students to choose (high school, college) over no high school. Note that changes in high school wages play a more important role when the likelihood of accessing higher education is lower – the opposite of what I observe for changes in y^C. In the extreme, when a student expects to access higher education with certainty ($\pi_c = 1$), changes in high school wages will not affect high school completion decisions through the education opportunities channel. In this path, high school earnings are viewed as the "consolation prize" a student gets if they fail to graduate from higher education, and the weight given to high school earnings is proportional to how likely a student perceives that failure to be.

Net effect.

$$\frac{\partial S^*}{\partial y^{HS}} > 0. \tag{17}$$

Increases in high school wages unambiguously raise high school completion through both margins: movements from (no high school, no college) to either (high school, no college) or (high school, college).

3.3.5 Primary-educated wages

Similar to high school wages, changes in primary-level wages will affect high school completion through both transitions.

Effect via transitions from (no high school, no college) to (high school, no college). Using the product rule on $\bar{p}_{10}^{H} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})R$:

$$\frac{\partial \bar{p}_{10}^{H}}{\partial y^{NH}} = -R\left[3 + (\bar{k}_1 + y_1^{NH})\left(\frac{4\beta_2}{\bar{k}_2 + y_2^{NH}} + \frac{\beta_3}{\bar{k} + y^{NH}}\right)\right] < 0. \tag{18}$$

Effect via transitions from (no high school, no college) to (high school, college). y^{NH} enters $V(0,0)$ through $\ln(\bar{k}_1 + y_1^{NH})$, $\ln(\bar{k}_2 + y_2^{NH})$, and $\ln(\bar{k} + y^{NH})$:

$$\frac{\partial\big[V(1,1) - V(0,0)\big]}{\partial y^{NH}} = -\frac{3}{\bar{k}_1 + y_1^{NH}} - \frac{4\beta_2}{\bar{k}_2 + y_2^{NH}} - \frac{\beta_3}{\bar{k} + y^{NH}} < 0, \tag{19}$$

so $\partial \bar{p}_{11|00}^{c}(p^H)/\partial y^{NH} < 0$ for all p^H.

Net effect.

$$\frac{\partial S^*}{\partial y^{NH}} < 0. \tag{20}$$

Higher dropout wages unambiguously reduce the share of students who graduate from high school by making the dropout option more attractive and the foregone earnings more costly, thereby steering students away from both paths leading to high school graduation.

4 Data and sample characteristics

To obtain data on college openings, I construct a novel dataset using institutional documents and information provided on college websites. This dataset contains the opening year, the year authorization to operate was granted, municipality name, region and state, and the type of higher education institution (university or trade school) that opened as a part of the expansion program, ranging from 2003 to 2014. I define the treatment year as the year an HEI opened, and, in cases where this information is missing (15%), I replace it with the year authorization to operate was granted. I complement these data with the Higher Education Census data for 2002 and 2018, provided by the Ministry of Education, to obtain information on the number of HEIs pre- and

post-policy and to determine whether receiving municipalities had a public or private institution before 2003.

For all of my analyses, I construct a panel dataset of municipalities, covering 2007 to 2019 for 9th-grade test scores, dropout rates from 9th to 12th grade, high school progression and graduation rates, and labor market outcomes, and 2010 to 2017 for college-going outcomes (share of students taking a college-entry exam, exam scores, and share who applied, were accepted and enrolled in an HEI). The year restrictions are due to data availability. For all of my pre-tertiary education and college-going outcomes, I restrict my sample to public school students at baseline, but allow them to move to private high schools as a possible response to the policy. This is a key sample restriction, since public school students, on average, are lower-income and less likely to migrate to pursue tertiary education outside their municipality, and would likely be the most affected by this policy.

For 9th-grade standardized test scores, I use data from the SAEB/Prova Brasil, a national test administered every two years. Test scores are available biennially from 1995 to 2019, but municipality identifiers are available only from 2007 onward. I thus restrict my test score analysis to municipalities treated between 2008 and 2014. For dropout rates, I use annual municipality-level rates provided by the School Census. To match the sample for test scores and high school graduation, I restrict my dropout rates analysis to municipalities treated between 2008 and 2014. For graduation and high school transition outcomes, I use the universe of K-12 student enrollment in Brazil from the 2007 to 2017 School Census. I build high school graduation outcomes from both 9th- and 10th-grade baselines: conditional on having enrolled in 9th/10th grade, does a student graduate on time or with up to a 2-year delay? High school transition uses the 9th grade baseline (last grade of primary school): conditional on enrolling in 9th grade, does the student enroll in 10th grade within 3 years? I then generate graduation and transition rates at the municipality level. My main outcome, high school completion rates, includes municipalities treated between 2008 and 2013, and high school transition rates include municipalities treated between 2008 and 2014.

My preferred graduation outcome uses the 10th-grade baseline for a few reasons. First, to construct graduation outcomes from the enrollment data, I need to observe students long enough into the future to determine whether they have completed 12th grade. Restricting to a 10th-grade baseline allows me to keep as much of the sample as possible while still restricting my baseline to a period prior to a critical dropout point (between grades 10 and 11) in the Brazilian system. This choice is further supported by the measured effect of the policy on dropout rates: I find no

changes in 9th-grade dropout rates, but a decrease in dropout at 10th grade. Second, conditioning on students who have already started their first year of high school (10th grade in Brazil) implies that, for those students, supply-side constraints are unlikely to be binding. Therefore, it follows that dropping out of or graduating from high school is mostly mediated by demand-side factors, which are the focus of this paper. For robustness, I present high school completion estimates using the 9th-grade baseline, which are similar to my preferred estimates.

For both 9th- and 10th-grade baseline graduation outcomes, a limitation of using enrollment data is that I must infer that students who reach 12th grade and then disappear from the K-12 enrollment system have graduated. To bound my estimates, I use 12th-grade yearly municipality dropout rates and present them in Table A1 in the Appendix. The results do not change, and, in some cases, the point estimates are larger. To avoid relying on imputed data, I use the outcome constructed solely from enrollment data as my main outcome.

College-going outcomes stem from the unified selection system, SISU, which has been widely used for admissions to public HEIs since 2010. This contains individual-level information on the applicant's college-entry exam scores, municipality of residence, school municipality, whether they were accepted into college using SISU, and whether they enrolled. I aggregate this information to generate municipality-level mean test scores, share of 17- to 25-year-olds taking the national college entry exam, and application, acceptance, and enrollment rates from 2010 to 2017. It is worth noting that not all HEIs use the national exam and SISU for admission. Thus, the acceptance and enrollment rates calculated from this dataset reflect college-going outcomes only for a specific subset of HEIs. However, nearly all schools opening as part of the expansion policy use SISU as their only form of admission, and these outcomes are therefore informative of whether students attend HEIs created by this policy. Due to data availability, I restrict my analysis of college-going to municipalities that received an HEI between 2011 and 2014.

To analyze possible changes in the local labor market, both in the short- and long-run, I employ administrative data from the Relação Anual de Informações Sociais (RAIS - Annual Report of Social Information), which contains the universe of formal sector employees and provides information on wages and hours worked, along with demographic characteristics such as gender, race, and educational attainment, and spans 2007 to 2019.

Throughout all of my estimates, I use data on municipality GDP per capita, the share of the agricultural sector in GDP, per capita education expenditure, Human Development Index (HDI), whether the local government is aligned with the federal government, the existence of an HEI

pre-policy, and population size as pre-policy municipality-level controls. These data come from the IPEA Data database and are discretized into quintiles. In all of my estimations, I restrict my sample to treated municipalities and to controls located at least 45 kilometers from a treated municipality.

4.1 Placement of HEIs

Placement of HEIs in this policy was not random. The federal government followed several criteria to select receiving municipalities, so treated units may differ from control units in many ways. Table 1 presents descriptive statistics on municipality characteristics, students' demographic characteristics, and pre-tertiary outcomes at baseline for treated and control municipalities, as well as for those receiving a trade school or a university.

Table 1: Pretreatment Characteristics by Treatment Status and Type of Treatment Received

	Treated (1)	Control (2)	P-value	Trade Schools (4)	Universities (5)	P-value
Municipality Characteristics						
GDP per capita (R$)	6,428.41	4,682.67	0.00	6,403.90	6,161.99	0.18
Share of Agriculture in GDP	0.13	0.29	0.00	0.12	0.16	0.00
HDI	0.57	0.50	0.00	0.57	0.55	0.00
Mayor aligned with federal government	0.09	0.08	0.09	0.09	0.09	0.86
Education spending per capita (R)	138.10	238.16	0.00	137.54	155.33	0.00
Had an HEI before the policy	0.42	0.05	0.00	0.43	0.30	0.00
High Secondary Premium	0.58	0.51	0.02	0.58	0.57	0.88
High College Premium	0.62	0.55	0.02	0.62	0.63	0.86
Number of Bolsa Familia Families (per capita)	0.07	0.09	0.00	0.07	0.08	0.22
Student's Characteristics - 9th Grade						
White	0.35	0.34	0.70	0.35	0.34	0.67
Working	0.22	0.27	0.00	0.21	0.23	0.07
Female	0.55	0.55	0.71	0.55	0.56	0.04
Mother completed High School	0.31	0.26	0.00	0.31	0.31	0.73
Mother completed higher education	0.09	0.09	0.20	0.09	0.09	0.94
Outcomes at baseline						
Portuguese Scores - 9th Grade	-0.39	-0.47	0.00	-0.38	-0.40	0.55
Math Scores - 9th Grade	-0.17	-0.21	0.01	-0.16	-0.19	0.52
High School Graduation	0.55	0.59	0.00	0.55	0.53	0.09
Dropout Rate - 9th Grade	7.57	7.81	0.50	7.15	8.63	0.08
Dropout Rate - 10th Grade	18.21	16.35	0.00	17.70	19.01	0.33
Dropout Rate - 11th Grade	14.31	12.15	0.00	13.97	14.54	0.61
Dropout Rate - 12th Grade	10.62	8.28	0.00	10.43	9.99	0.64
Observations	379	2,180		305	46	

Note: Test scores are z-scores standardized with respect to the population mean for each year, which includes private schools in Brazil and, thus, is the reason why public schools' test scores for both treated and control groups are negative. High School Graduation is the share of students entering 10th grade in 2007 who completed high school within five years. Dropout rates are the 2007 dropout rates in urban public schools for each grade. 28 municipalities received both a university and a trade school, and are therefore left out of the trade school vs university breakdown.

Overall, treated municipalities have higher GDP per capita, are less agriculture-intensive, have a higher HDI, and lower K-12 education spending than control municipalities. Contrary to the federal government's stated criteria, 42% of treated municipalities also had a pre-existing HEI, compared with only 5% in the control group. They are also more likely to be classified as a high secondary and college premium municipality, and have fewer families enrolled in the conditional cash transfer program, Bolsa Familia. Treated municipalities have a lower share of students who reported working outside of the household and a higher share of students whose mothers had completed high school, although a similar share of students had mothers who completed higher education. Public school 9th-graders in treated municipalities have higher Portuguese and Math test scores than those in the control group. However, both groups are below the national average on the standardized tests, a result of the public school student sample restriction. On the other hand, conditional on starting 10th grade, students in control municipalities were 4 percentage points more likely to graduate from high school. Treated municipalities also had higher dropout rates at all three grades of high school.

Since municipality characteristics may vary by type of treatment received, columns 4 and 5 of Table 1 compare pretreatment characteristics of municipalities that received a university versus those that received a trade school. The latter are less agriculture-intensive and have slightly higher HDI scores than those receiving a university. I do not find large differences in students' characteristics or pre-treatment outcomes between municipalities that received a trade school and those that received a university.

Table 2: Correlation Between Pre-Treatment Outcomes and Treatment Status

	Independent Variable (2007 level)						
				Dropout rate			
	Graduate on time	Eventually graduate	Start HS	9th grade	10th grade	11th grade	12th grade
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent Variable							
Treated	-1.005***	-0.770**	0.552	-0.017**	0.007	0.012**	0.008
	(0.370)	(0.374)	(0.464)	(0.008)	(0.004)	(0.005)	(0.006)
Observations	2,400	2,400	2,400	2,525	2,525	2,525	2,525

Note: Each column reports the probit coefficient from a regression of treatment status on the 2007 level of the outcome in the column header. The sample includes one observation per municipality: municipalities that received their first public HEI between 2008 and 2014 and the never-treated comparison municipalities used in the main results. Graduation outcomes refer to the 2007 10th-grade cohort; Start HS refers to the 2007 9th-grade cohort. All specifications include the same municipality-level control variables as the main results, measured pre-policy and discretized. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

One possible concern is that treated localities were chosen based on pre-treatment outcomes.

For instance, municipalities with high secondary graduation rates pre-policy could have received new HEIs at higher rates due to increased demand for tertiary education. To assess if this is the case, I analyze whether pre-treatment high school graduation, transition and dropout rates predict treatment assignment by running a probit regression of treatment status on pre-policy outcomes and municipality-level controls used in the main analysis. Estimates are presented in Table 2. Although a lower 9th-grade dropout rate is correlated with receiving a new HEI, worse outcomes at high school predict treatment: treated municipalities had lower high school completion and higher 11th-grade dropout rates pre-policy. If selection bias exists, it would bias my estimates downward and thus should not be a concern. This relationship is also expected, given the policy's stated goal of serving underserved and underdeveloped regions. To assess differences in trends, I will present event-study estimates in the main results section.

I will deal with the observed differences between treated and control municipalities in two ways. First, in a difference-in-differences framework, control and treated units need to be comparable in trends but not in levels. I will show in the results section that there is little evidence of pre-trends, which supports the comparisons between these two groups. Second, I will control for an array of municipality characteristics that might have played a role in the federal government selection criteria. I expand these to include a political alignment dummy that identifies whether the local government was aligned with the federal government, following Arroyo and Petterini (2020), and per-capita educational expenditure to account for potential differences in pre-treatment school quality. As a robustness check, I include the pre-policy number of families receiving the conditional cash transfer program, Bolsa Familia. This is relevant because the program's conditionality requires students to be enrolled in school. The results do not change when including this additional control.

5 Methods

To estimate the effects of higher education expansion on local students' pre-tertiary education and college-going outcomes, I will compare treated versus never-treated municipalities before and after an HEI's opening in a difference-in-differences setting. Equation 21 presents a two-way-fixed-effect estimation for that comparison.

$$y_{m,t} = \alpha + \beta ReceivedHEI_m \cdot Post_{t,m} + \gamma_t + \mu_m + \Pi \mathbf{X}_m + \epsilon_{m,t} \tag{21}$$

Where $y_{m,t}$ is the outcome of interest in municipality m at time t; $ReceivedHEI_m$ is the

treatment variable, which equals one if the municipality received a higher education institution; $Post_{t,m}$ equals one if the period t is post-treatment in municipality m and zero otherwise; γ_t are year fixed effects to control for national trends in outcomes; μ_m are municipality fixed effects; and $\mathbf{X}$ is a vector of discretized pre-treatment municipality characteristics, which controls for factors taken into account by the Federal Government to assign treatment[7]. The main coefficient of interest is β, which in this specification does not vary with time.

Since the policy is staggered and the treatment years in my analysis range from 2008 to 2014, I employ the staggered-timing difference-in-differences estimator proposed by Callaway and Sant'Anna (2021) and include municipality-level controls. This will allow for heterogeneous effects over time that should be relevant to this policy, as well as mitigate concerns about employing a Two-Way Fixed Effects estimator in a staggered treatment setting (Goodman-Bacon, 2021). In this approach, the inverse-probability-weighted (IPW) version of the average treatment effect on the treated is given by Equation 22.

$$ATT(g,t) = E\left[\left(\frac{G_g}{\mathbb{E}[G_g]} - \frac{\frac{p_g(X)(1-G_g)}{1-p_g(X)}}{\mathbb{E}\left[\frac{p_g(X)(1-G_g)}{1-p_g(X)}\right]}\right)\cdot(Y_t - Y_{g-1})\right] \tag{22}$$

where $ATT(g,t)$ is the average treatment effect measured at time t, $t \geq g$, for all municipalities that received an HEI starting in time g. $G_g = 1$ if a municipality is part of the group g, Y_t is the outcome for all units in the post-period of interest, t, and Y_{g-1} is the outcome for all units in the period immediately before when treatment started, $g-1$. $p_g(X) = P(G = g \mid X, G \in \{g, \text{control}\})$ is a propensity score calculated using a set of municipality-level covariates measured pre-policy that captures the likelihood of belonging to cohort g conditional on covariates X and being either in cohort g or the control group. $\mathbb{E}[G_g]$ and $\mathbb{E}\left[\frac{p_g(X)(1-G_g)}{1-p_g(X)}\right]$ are normalizations to ensure that weights sum to one for both treated and control groups. In practice, I use the doubly robust (DR-IPW) version of this estimator, which additionally subtracts from $(Y_t - Y_{g-1})$ the change in the outcome predicted for control municipalities with the same pre-policy characteristics, $m_{g,t}(X)$, and is consistent if either the propensity score or the outcome model is correctly specified (Sant'Anna and Zhao, 2020)

ATTs for each time t and treated cohort g are then grouped following Equation 23, using the

[7]Pre-tertiary outcomes are the likelihood of transitioning into high school, high school graduation, 9th grade Portuguese and Math scores, and average college-entry test scores. College-going outcomes are the share of 17- to 25-year-olds who take the college-entry exam, apply to college, are accepted, and enroll. Labor market outcomes are log wages by educational level.

weighting function $w(g,t)$. This will allow me to estimate an overall average effect for the policy, θ.

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} w(g,t) \cdot ATT(g,t) \tag{23}$$

My main identification assumption is that, in the absence of treatment, treated and untreated municipalities would have followed the same trends in the outcomes over time. With the inclusion of controls, the parallel trends assumption needs to hold conditional on municipality characteristics. Although this hypothesis cannot be formally tested, I will use event-study specifications as described in Equation 24 to provide evidence of no differential pre-trends. This specification will allow for the aggregation of ATTs across all treated cohorts, g, for each pre and post-policy period, ℓ. When $\ell < g$, the estimated ATT works as a placebo test that can provide suggestive evidence of whether the parallel trends assumption is met. Control municipalities are at least 45 kilometers away from a treated municipality and are never treated. Municipalities treated before the period for which data are available are excluded from my analysis.

$$\theta(\ell) = \sum_{g \in G} w(g,\ell)\, ATT(g, g+\ell), \quad \text{for each event time } \ell. \tag{24}$$

6 The impacts of higher education opportunities on earlier educational decisions

Table 3 shows the effects of higher education expansion on pre-tertiary outcomes, using Callaway and Sant'Anna (2021) and measured within 3 years of an HEI's opening. The opening of a new HEI in a student's municipality does not affect the likelihood that 9th-graders transition to high school in the following three years or 9th-grade dropout rates. However, persistence increases in all high school grades: 10th graders are 1.2 percentage points less likely to drop out, while dropout rates for 11th and 12th graders decrease by 0.8 and 0.6 percentage points, respectively. These effects represent a 5–6% decrease relative to the baseline mean. Figure 4 shows the timing of these effects. While 10th-grade dropout decreases in the period immediately after a new campus opens, the effects in 11th and 12th grade appear only two years after treatment. Tenth graders are thus the most responsive to new college opportunities.

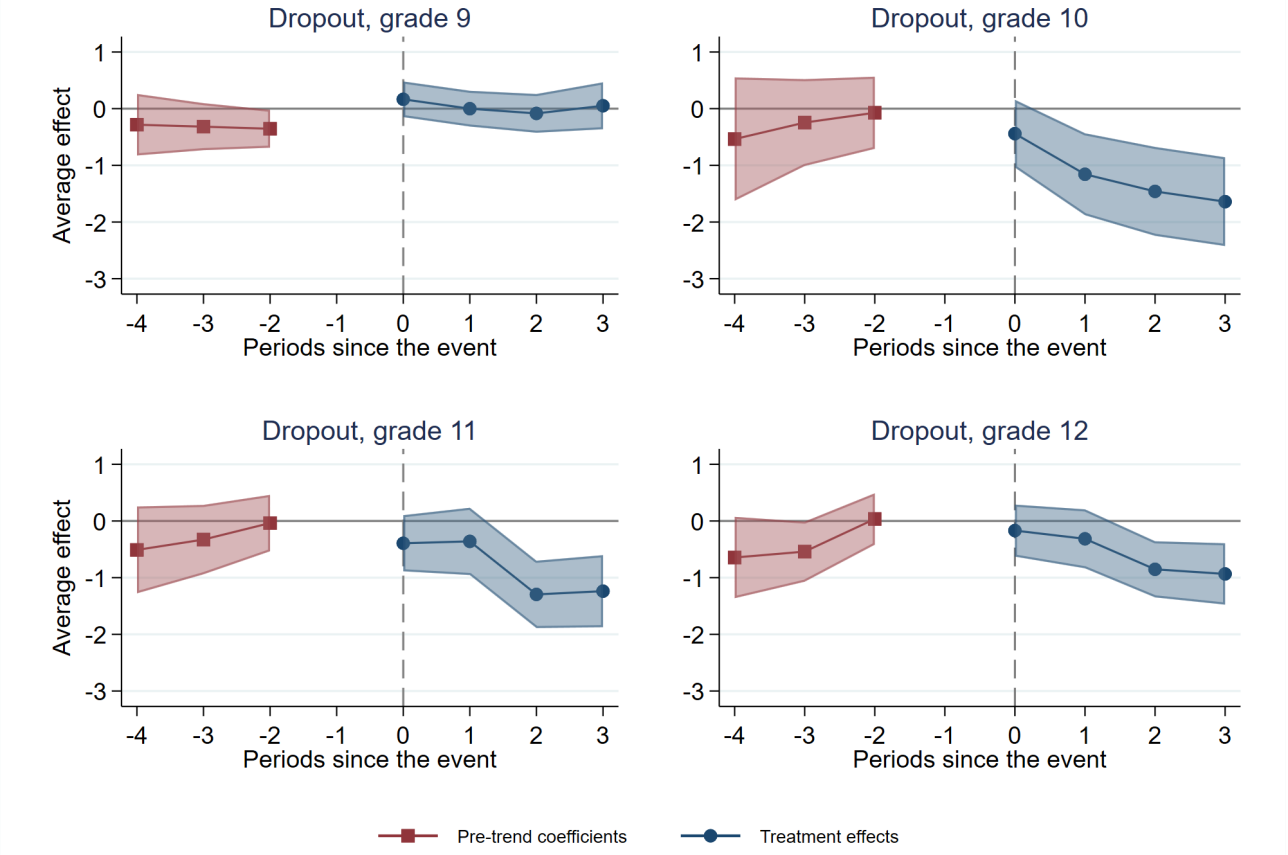
Given that students increase persistence in all grades of high school following the opening of an

Table 3: Persistence in Secondary School

		Dropout rate					
	Start HS	9th grade	10th grade	11th grade	12th grade	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated	0.005*	0.034	-1.174***	-0.822***	-0.568***	-0.003	0.008**
	(0.003)	(0.149)	(0.304)	(0.251)	(0.219)	(0.003)	(0.003)
Observations	19,208	35,241	35,241	35,241	35,241	21,616	16,814
Pre-policy Mean	0.79	7.57	18.21	14.31	10.62	0.46	0.54

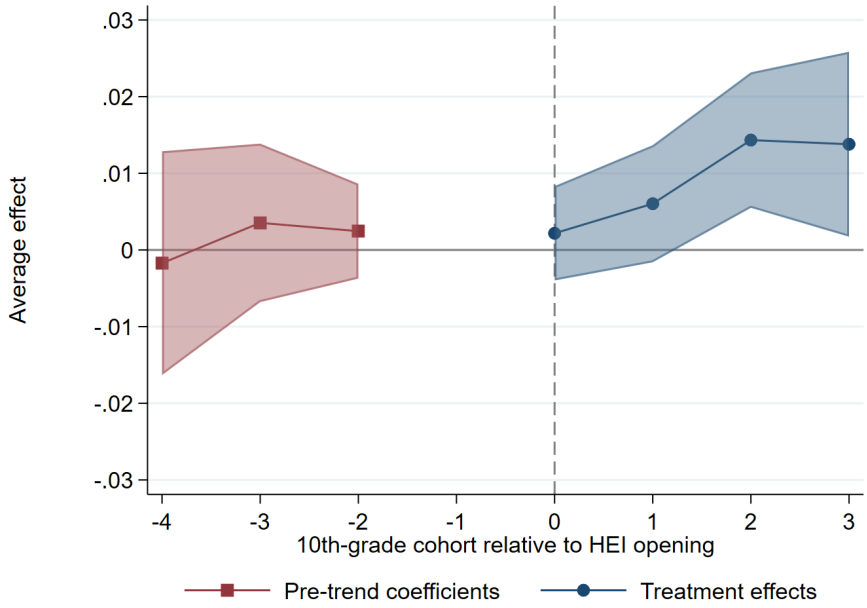
Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. Dropout estimates (columns 2–5) use a common sample of municipality-years with non-missing dropout rates for all four grades. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 4: Effect of new HEIs on dropout rates



HEI, I then assess whether this translates to an increase in the likelihood of completion. Columns 6 and 7 show effects on high school completion for students who were in 10th grade at the time

Figure 5: Likelihood of graduating High School with up to a 2-year delay



of treatment, and summarize the effects for the first three exposed cohorts. Treated students are not more or less likely to graduate from high school on time. However, they are 1 percentage point more likely to graduate with up to a 2-year delay, which corresponds to a 2% increase relative to the pre-policy mean. This corresponds to over 20,000 additional high school graduates in the first three exposed cohorts following the policy. Figure 5 shows that the effects on high school graduation increase over time, similarly to the observed pattern for dropout rates at 10th grade.

Table 4 presents performance outcomes for 9th-graders and for students exiting high school who choose to take the college-entry exam. I find no changes in either 9th-graders' Portuguese and Math performance or college-entry exam scores. I do, however, find meaningful increases in the share of local young adults taking an optional college exam. The share of students writing the exam following the opening of an HEI increases by 2 percentage points, a 20% increase from the pre-policy mean. This increase in take-up does not seem to come at the expense of performance, as measured by college-entry exam scores.

6.1 Does the type of Higher Education matter?

The positive effects above are comparable in magnitude with those in the Indian setting, as measured by Jagnani and Khanna (2020). Unlike that setting, however, the Brazilian policy not

Table 4: Performance Outcomes

	9th Grade Math	9th Grade Portuguese	College Exam Scores	Share Taking Exam
	(1)	(2)	(3)	(4)
Treated	0.01	0.01	-0.62	0.02***
	(0.01)	(0.01)	(1.38)	(0.00)
Observations	17,548	17,548	10,576	10,576
Pre-policy Mean	-0.17	-0.39	525.48	0.10

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

only expanded existing 4-year universities but also established new 4-year higher education institutions and 2-year trade schools. This policy's unique feature allows for testing whether other types of higher education opportunities can shift pre-tertiary students' schooling decisions. In this section, I examine whether the average effects measured above are driven by a specific type of HEI.

Although not statistically different, 9th graders in municipalities that received a university are 2 percentage points more likely to start high school in subsequent years, with no effect for students in localities that received a trade school. Although trade schools do not seem to drive students to start high school in the first place, they meaningfully change persistence in high school, decreasing dropout rates at all grades and increasing high school graduation.

This suggests that the opportunity to access a trade school is more salient to students later in their schooling and who would have started high school in the absence of the policy. University openings, on the other hand, might change the composition of who starts high school in the first place. This difference in the composition of high school is further evidenced when comparing the 9th- and 10th-grade baseline estimates of the likelihood of eventually graduating, Table A3 in the Appendix. When conditioning on 9th-grade enrollment, I find a large 4% increase in the likelihood of eventually completing high school for students exposed to a university, although only marginally significant and not statistically different from the effect for trade schools. This effect completely dissipates if conditioning on enrollment in 10th grade, likely due to changes in high school composition driven by the newly opened university.

Table 5: Persistence in Secondary School, by Type of HEI Received

		Dropout rate					
	Start HS	9th grade	10th grade	11th grade	12th grade	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Received a Trade School	0.003	0.038	-1.239***	-0.760***	-0.584**	-0.004	0.010**
	(0.003)	(0.159)	(0.333)	(0.277)	(0.238)	(0.004)	(0.004)
Observations	18,616	34,149	34,149	34,149	34,149	20,950	16,296
Pre-policy Mean	0.79	7.15	17.70	13.97	10.43	0.47	0.55
Received a University	0.016*	0.225	-0.803	-0.949	-0.321	-0.007	-0.003
	(0.009)	(0.436)	(0.739)	(0.593)	(0.563)	(0.008)	(0.010)
Observations	16,544	30,355	30,355	30,355	30,355	18,619	14,483
Pre-policy Mean	0.78	8.63	19.01	14.54	9.99	0.44	0.53
P-value (diff)	0.18	0.69	0.59	0.77	0.67	0.74	0.21
Bootstrap p-value (diff)	0.18	0.70	0.60	0.77	0.66	0.77	0.23

Note: Standard errors in parentheses, clustered at the municipality level. Each type of HEI is compared with the same never-treated municipalities. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). P-value (diff) tests the equality of the trade school and university estimates using $\sqrt{se_{TS}^2 + se_U^2}$, which ignores that both groups share the same comparison municipalities. Bootstrap p-value (diff) uses the standard error of the difference from a municipality-level bootstrap (200 replications, stratified by opening cohort), re-estimating both ATTs in each draw. Observations vary between outcomes due to years of data availability. Dropout estimates (columns 2–5) use a common sample of municipality-years with non-missing dropout rates for all four grades. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

6.2 Who benefits from higher education expansion?

Since all students in my sample are public school students, the overall effects are informative of how the policy affects lower-income, highly constrained students. However, within public schools, some students are more disadvantaged than others. In this section, I analyze whether the policy had heterogeneous effects across racial groups. Pre-policy, non-white students are 6 percentage points less likely to ever enroll in high school (70% versus 76% for white students) and 6 percentage points less likely to ever graduate (53% compared to 59% for white students).

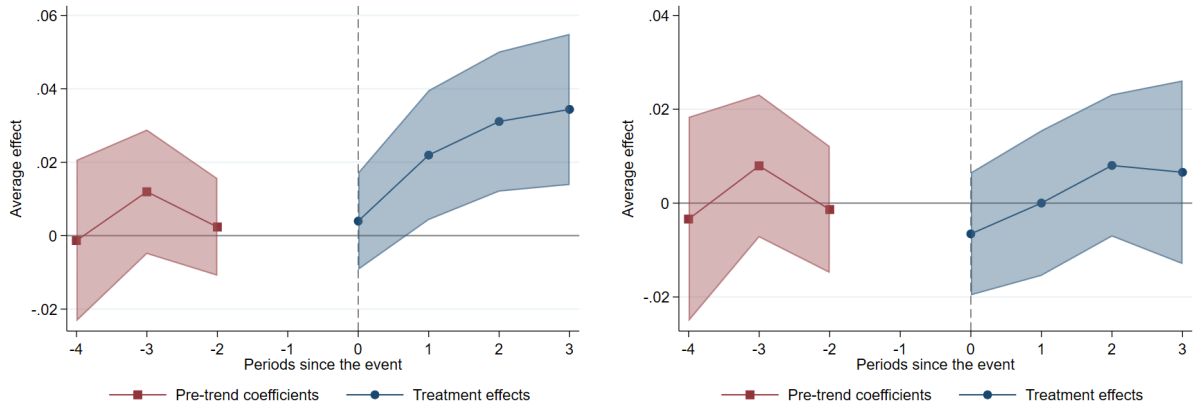
Table 6 shows that the policy disproportionately benefits non-white students. They are 2 percentage points more likely to ever enroll in high school (a 3% increase from the pre-policy mean), closing 1/3 of the pre-policy gap between white and non-white students. Although only marginally statistically different, estimates on the likelihood of ever graduating high school for non-white students are four times as large as those for white students: conditional on being enrolled in 10th grade, non-white students are 2 percentage points more likely to eventually graduate high school, a 4% increase from the pre-policy mean and 1/3 of the pre-policy gap relative to white students. Figure 6 shows the event studies for the share of non-white and white students starting high school following the policy. The effect is increasing over time for non-white students, and flat for white students.

Table 6: Persistence in Secondary School, by Race

	Start HS (1)	Graduate on time (2)	Eventually graduate (3)
White	0.001	-0.006	0.005
	(0.007)	(0.007)	(0.008)
Observations	17,529	20,413	15,807
Pre-policy Mean	0.76	0.50	0.59
Non-white	0.021***	0.004	0.020***
	(0.007)	(0.006)	(0.007)
Observations	17,529	20,413	15,807
Pre-policy Mean	0.70	0.42	0.53
P-value (diff)	0.05	0.27	0.18
Bootstrap p-value (diff)	0.02	0.13	0.08

Note: Standard errors in parentheses, clustered at the municipality level. Estimates for white and non-white students are obtained in separate regressions that use the same treated and never-treated municipalities, and each column keeps the same municipality-years for both groups. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). P-value (diff) tests the equality of the two estimates using $\sqrt{se_1^2 + se_2^2}$, which ignores that both groups come from the same municipalities. Bootstrap p-value (diff) uses the standard error of the difference from a municipality-level bootstrap (200 replications, stratified by opening cohort), re-estimating both ATTs in each draw. Observations vary between outcomes due to years of data availability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 6: Share of non-white (left) and white (right) 9th Graders who Start High School



7 Channels of impact: how does higher education expansion affect pre-tertiary educational attainment?

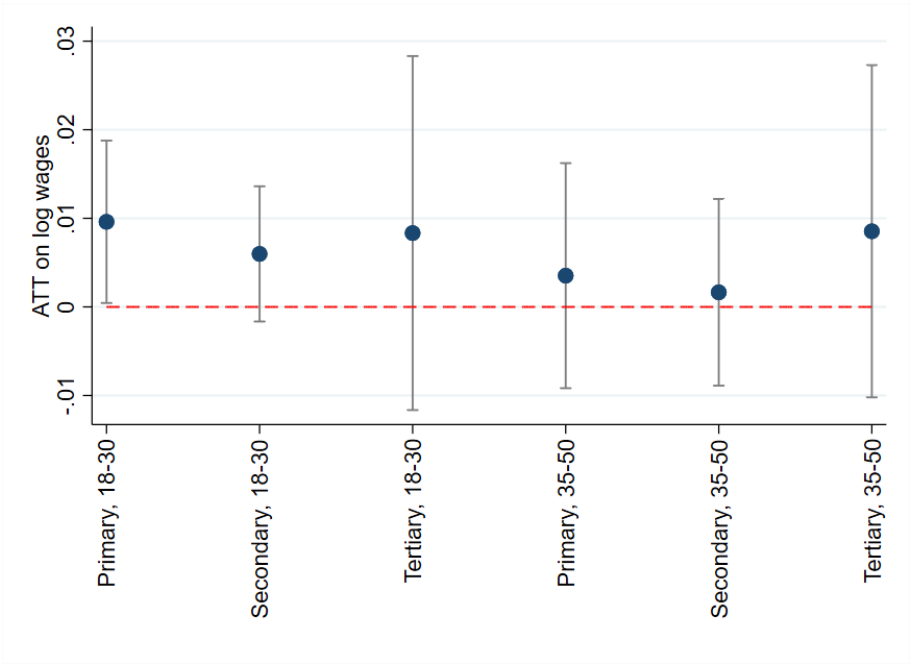
As highlighted in the model, changes in high school completion following the opening of an HEI can stem from two channels: the direct education opportunities channel, through changes in college attendance costs, p^C, and the likelihood of higher education admission, π_c; and the indirect labor

market channel, through changes in local wages arising from general equilibrium effects and new employment opportunities. In this section, I will first provide evidence on the mechanisms driving the observed effects by empirically examining which parameters change as a result of this policy. I will then empirically assess whether the necessary conditions derived in Section 3.2 hold.

7.1 The indirect labor market channel: changes in local wages

As documented in the literature (Jagnani and Khanna, 2020; Vu, 2024; Cox, 2024; Feinmann and Hsu Rocha, 2026; Lallemant and Dillon, 2026), the arrival of new HEIs can affect the local economy in many ways, such as by changing the available infrastructure and employment opportunities. New HEIs can then affect local wage levels by changing the local labor market, which can in turn affect high school completion as predicted by my framework. As derived in the comparative statics section, an increase in high school or college earnings, and a decrease in primary education wages could all be consistent with an increase in high school graduation following the opening of an HEI. In this section, I assess whether the policy affected wages in the formal labor market in the short run (within 3 years of an HEI's opening) across different education levels and ages.

Figure 7: Effect on formal-sector wages by level of education and age



Changes in the local labor market do not seem to be driving the observed increases in high school completion. Point estimates are largest and significant only for primary-educated earnings

among young workers, although all estimates shown in Figure 7 are indistinguishable from one another. These increases are likely a result of young individuals choosing to stay in school longer as a result of the policy, thus decreasing the supply of low-skilled labor in the short-run. If primary wages are, in fact, increasing as a result of the policy, the framework predicts that dropouts should increase, and high school graduation rates would move in the opposite direction of the observed effects. Therefore, if the labor market channel is at play, it operates by dampening the policy's positive effects on pre-tertiary educational attainment rather than driving them.

Since labor market changes are the only channels through which movements from (no high school, no college) to (high school, no college) occur, it must be true that the observed positive effects stem solely from transitions directly into the (high school, college) path. The marginal students driving the results are those who would otherwise never graduate from high school but choose to do so because of increased higher-education opportunities.

7.2 The direct education opportunities channel

7.2.1 Probability of attending higher education

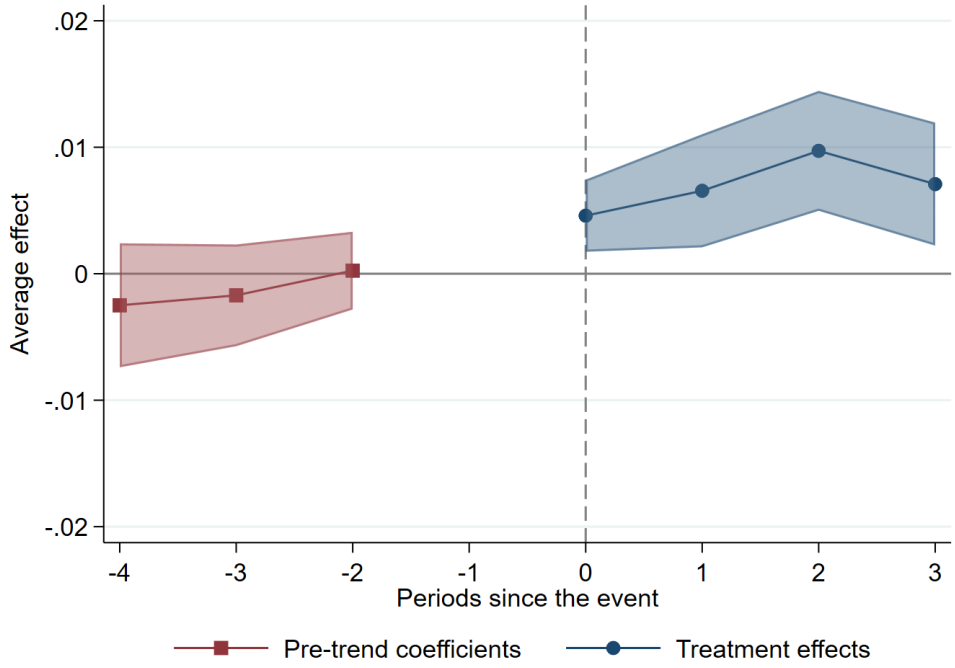
Although an increase in the supply of higher education is likely to raise a local student's probability of attending the new local HEI, this effect is not guaranteed. The national college entrance exam and unified higher education application system allow higher-income students from other municipalities to migrate and take advantage of newly created spots. In addition, spots at federal tuition-free HEIs are limited, and acceptance rates often range from 5 to 15%, since the application is nearly costless. To assess whether the policy increases higher education attendance for local public school students, I estimate the effect of receiving an HEI on the likelihood that local students apply to, are accepted to, and enroll in federal HEIs. Table 7 shows that local students are 2.5 percentage points more likely to apply (14% increase relative to the pre-policy mean), 1.8 percentage points more likely to be accepted (nearly a 130% increase relative to the pre-policy mean), and 0.7 percentage points more likely to enroll (a 140% increase) in a public federal HEI following the policy. Enrollment effects are stable over time, as shown in Figure 8, confirming that local students experience a continued increased probability of attending higher education (as represented by π_c in my framework).

Table 7: College-Going Outcomes

	Applied (1)	Accepted (2)	Enrolled (3)
Treated	0.025***	0.018***	0.007***
	(0.007)	(0.005)	(0.002)
Observations	17,704	17,704	17,704
Pre-policy Mean	0.179	0.014	0.005

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 8: Likelihood of Enrolling in a Federal HEI



7.2.2 Tertiary education costs

Since public higher education institutions in Brazil are free, costs associated with higher education can be thought of as a function of proximity to a public HEI. By reducing the distance to a public HEI, this policy directly lowers the costs of attending higher education for treated students, who will no longer incur moving or significant transportation costs. As shown in the theoretical

framework section, if the policy reduces the cost of attending higher education, the share of students graduating high school should increase, all else constant. Although I cannot empirically measure the role of costs, it is reasonable to assume the policy is reducing costs related to higher-education attendance by reducing distance, as the relationship between distance to school and educational attainment has been widely documented in the literature (Card, 1993; Currie and Moretti, 2003; Carneiro et al., 2011; Muralidharan and Prakash, 2017).

7.3 The role of returns to education

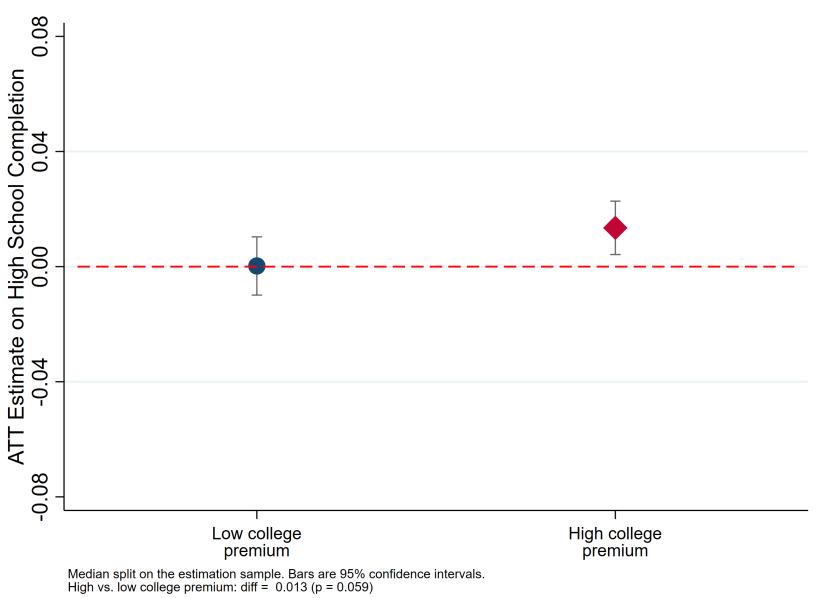
As derived in the framework, the relative magnitudes of local college and high school premiums determine whether an expansion in higher-education opportunities can affect the high school decision. For students to choose to graduate high school *because* of new college opportunities, college premiums need to be high enough that the possibility of completing college meaningfully raises the expected returns to high school. Additionally, for at least some students to drop out of high school in the absence of college, the additional return to graduating high school over primary education cannot be excessively high.

These two conditions generate two testable predictions: first, new college opportunities should affect students' high school decisions only where the college premium is high. Second, this effect should be largest where the high school premium is low, that is, where college opportunities would raise expected returns to high school the most. I test these predictions by exploring the distribution of college and high school premiums across Brazil, pre-policy.

Using data from the 2000 Brazilian Census, I construct municipality-level pre-policy college and high school premiums, defined as $\log(\frac{y_m^C}{y_m^{HS}})$ and $\log(\frac{y_m^{HS}}{y_m^{NH}})$, respectively, where y_m^C, y_m^{HS} and y_m^{NH} are municipality-level average earnings of workers with higher education, secondary school and primary school degrees. I then classify municipalities as high (low) college and high school premium if they are above (below) the median of the pre-policy premium distribution. Figure 9 plots the average treatment effects of receiving an HEI on high school graduation rates for municipalities with low and high pre-policy college premiums. Consistent with the predictions of the framework, high school completion increases only in municipalities with high college premiums. This effect is positive, significant, and statistically different from the effect measured for students in municipalities with below-median college premiums at 10%.

I further divide municipalities into four groups, now combining both the baseline college and high school premiums: low college premium and low high school premium; low college premium and

Figure 9: High School Completion effect, by pre-policy distribution of the college premium

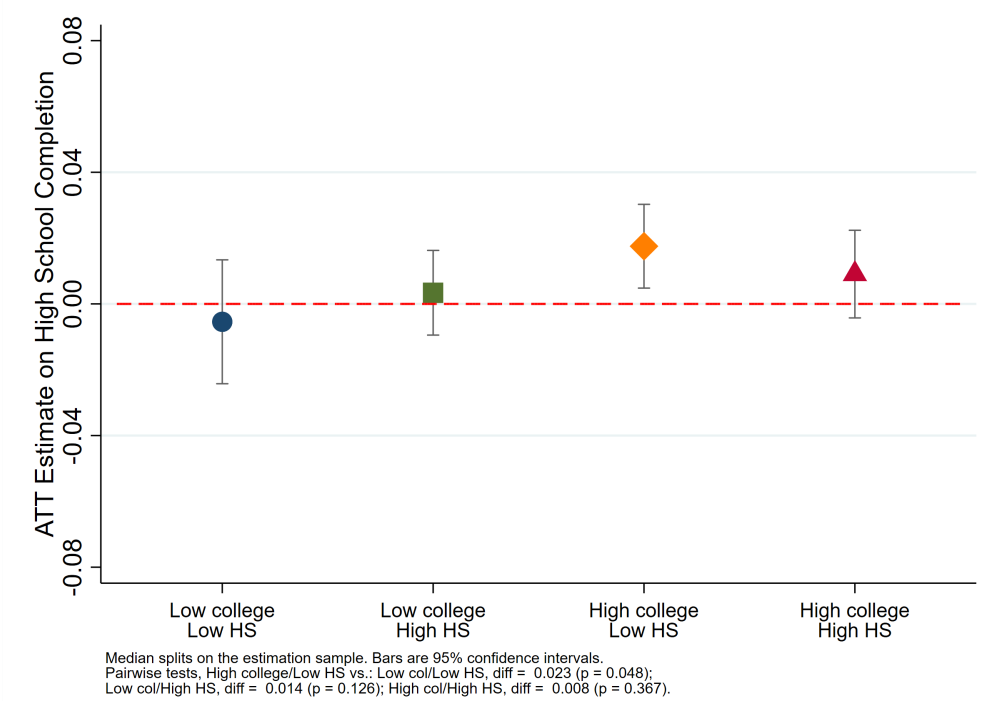


high high school premium; high college premium and low high school premium; and high college premium and high high school premium. Figure 10 plots the average treatment effects of receiving an HEI on high school graduation rates for each of these groups. Although not statistically different, within municipalities with high college premiums, point estimates are largest for those with low high school premium, as predicted by the framework.

These results suggest that the overall effects shown in Table 3 mask important heterogeneity in who responds to increased tertiary educational opportunities by changing pre-tertiary schooling decisions. This heterogeneity helps reconcile existing evidence in the literature: Kenya and India, like Brazil, have seen increases in pre-tertiary attainment following an expansion in higher education supply (Jagnani and Khanna, 2020; Erbabian, 2026), while the same is not observed for Chile and the United States (Bedard, 2001; Erbabian, 2026).

The first set of countries differs from the second in two key aspects that, as highlighted in the framework, can meaningfully shape high school responses to college opportunities. First, baseline high school completion is much higher in Chile and the United States, leaving little room for college opportunities to drive earlier educational attainment. Second, the structure of returns differs. As Gethin (2025) show, returns to secondary education in the United States exceed returns to primary education, while the additional return to tertiary education is a smaller jump than in Sub-Saharan Africa, India, and Brazil. Estimates in Montenegro and Patrinos (2014) show a similar pattern in

Figure 10: High School Completion effect, by pre-policy distribution of college and high school premium



Chile, where the return to secondary education exceeds that of primary education, so that, despite a large college premium, high school is already worth completing on its own. Taken together, my results and this cross-country evidence suggest that expanding college opportunities raises demand for earlier schooling where returns to high school alone are not enough to keep students in school, but may have little effect where returns to high school and graduation rates are already high.

8 How sustainable is the policy effect in the long run?

As students respond to this policy by increasing high school and college attainment, returns to both levels of education might decrease, diminishing the impact of higher education opportunities on high school completion over time. In this section, I provide suggestive evidence on how municipality-level college and high school premiums (in the formal labor market) behave following the opening of an HEI. This will inform on whether we can expect higher education expansion to permanently increase the returns to high school, or whether these short-run effects should dissipate over time.

Figure 11 shows that local college premiums remain stable for up to 6 years after an HEI's opening. On the other hand, high-school premiums begin to decrease sharply two years after the

Figure 11: Municipality college-premium (left) and high-school premium (right) in the formal labor market

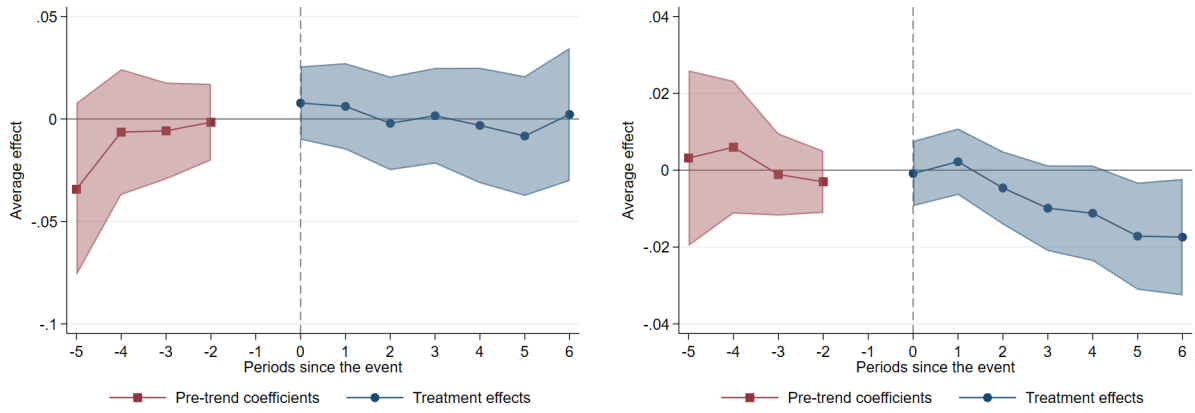
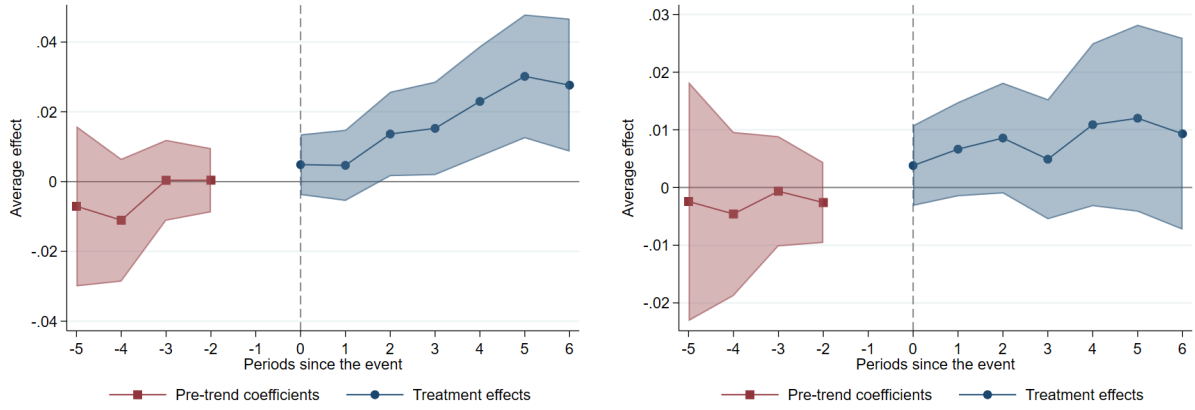


Figure 12: Primary education wages (left) and secondary education wages (right) in the formal labor market



policy. This decrease in high school returns comes from an increase in primary-education wages, rather than a decrease in earnings for high school graduates, as shown in Figure 12. This likely reflects both a reduced supply of primary-educated young workers due to increased educational attainment following the opening of an HEI and the potential creation of new jobs that might increase demand for low-skilled labor. Overall, college premiums seem to remain stable 6 years after the policy, suggesting that the higher-education-induced increase in returns to high school can be sustained over time, provided that college returns remain high relative to the outside option (in this case, primary education).

9 Robustness analysis

Changes in composition. One possible threat to identification is that, due to a potential influx of people into the municipality caused by the arrival of an HEI, such as professors, staff, and new students, the composition of public K-12 schools could change in ways that affect high school graduation rates and test scores. The observed results would then not be attributable to the expansion of higher education itself, but to compositional changes generated by the policy. To assess whether this is the case, I replace the outcome variable in my estimation with the shares of white students, students who attended only public schools throughout their lives, and students whose mothers completed high school. This will measure whether the policy affects student composition[8].

As shown in Table A7 in the Appendix, student composition does not appear to change as a result of the policy. In fact, all significant estimates would bias the results downward, given that non-white and public school students, on average, have lower high school completion rates and lower test scores than white and private school students.

Control variables. Results are robust to alternative sets and combinations of controls, including the number of families receiving Bolsa Família per capita. When controls are omitted entirely, estimates for dropout rates remain similar and are even larger, but the effect on high school completion is smaller (0.005 compared with 0.008 in the main specification) and no longer statistically significant. Because treated municipalities were selected based on observable characteristics that may also be associated with trends in educational outcomes, the identifying assumption in my main specification is parallel trends conditional on these pre-policy characteristics. The smaller unconditional estimate is consistent with this: without controls, treated municipalities are compared with control municipalities that differ systematically in characteristics such as income, population, and education spending. Estimates without controls are shown in Appendix Tables A4 and A5.

Permutation test. To test whether my results could arise by chance, I conduct a randomization-inference test for the effect on eventually graduating from high school (10th grade baseline). I randomly reassign the observed treatment cohorts, including never-treated status, across municipalities 2,000 times, holding fixed the number of municipalities in each cohort. In each draw, I re-estimate the ATT with the same specification as in the main results. Because the reassigned

[8]Public school attendance in Brazil is highly correlated with income. The questionnaire asks students if, starting in first grade, they studied in public only, private only, or both private and public schools.

cohorts are unrelated to where HEIs actually opened, these placebo estimates show the distribution of effects expected in the absence of any true policy effect. As expected, the placebo estimates are centered at zero (mean −0.0002). The observed t-statistic exceeds 98.7% of the placebo t-statistics in absolute value. It is therefore unlikely that my main results are driven by random noise. Figure A1 in the Appendix shows the distribution of placebo estimates.

Technical high schools. Although trade schools are part of the Brazilian higher education system, they also offer technical training to students enrolled in high school. To assess whether these programs drive the increase in high school completion, I re-estimate my main specification excluding students who pursue technical training during high school. As shown in Table A6 in the Appendix, coefficients are smaller and no longer statistically significant in this restricted sample, but they are not statistically different from the main estimates. Because take-up of technical training may itself respond to the opening of a trade school, this exercise should be interpreted with caution, as the restricted sample may differ between treated and control municipalities. Still, the results suggest that the effect on high school completion is not driven solely by students enrolled in technical programs during high school.

10 Discussion

Although future educational opportunities can meaningfully shape early educational attainment, this channel has only recently been explored in the literature and remains largely absent from the policy debate as a tool for sustainably increasing overall demand for schooling. By bringing higher education within reach of low-income students, expansion policies can increase pre-tertiary educational attainment and reduce educational disparities. Exploring a large-scale expansion of higher education institutions in Brazil, I find that local students exposed to a new HEI increase persistence in high school by 1 percentage point (5-6% relative to the baseline mean) and are 2% more likely to graduate high school – increasing the stock of high school graduates by over 20,000 in the first three exposed cohorts. More disadvantaged, non-white public students benefit the most: they are 3% more likely to start high school after an HEI opens and, among those who reach 10th grade, 4% more likely to graduate, closing one-third of the completion gap relative to white students.

Trade schools and universities both increase high school completion, but appear to affect students at different stages of their educational careers. Ninth graders in municipalities that received

a university were 2 percentage points more likely to transition to high school, while those in municipalities that received a trade school were not. Instead, trade schools shift schooling decisions among older pre-tertiary students who would have started high school in the absence of the policy, but are now more likely to stay in school and graduate. This result is particularly relevant for LMICs, since trade schools are smaller and less costly than four-year universities, and potentially better aligned with the needs of local labor markets.

My results stem from increased educational opportunities rather than from indirect labor market effects. I find that students are 14% more likely to apply to an HEI and experience a nearly 1-percentage-point increase in the likelihood of enrolling (a 140% increase from the pre-policy mean). On the other hand, labor market changes cannot explain my findings: earnings of young primary-educated workers rise after an HEI opens, which, if anything, makes dropping out more attractive and would push my estimates in the opposite direction.

A key prediction of the theoretical framework is that higher education opportunities should only matter if college premiums are high relative to high school returns. I find empirical evidence to support this prediction. Only in municipalities with a high college premium pre-policy do students respond to a new HEI by increasing secondary schooling completion. This effect is largest in municipalities where the college premium is high and the high school premium is low, highlighting the important role that future educational opportunities play in making pre-tertiary education more worthwhile.

Last, I examine whether these results can be sustained over time by analyzing how the college and high school premiums in the formal labor market evolve up to six years after an HEI opens. While college premiums remain stable over time, high school premiums decline sharply as wages for primary-educated workers rise. This is consistent with general equilibrium effects of expanding education documented by Khanna (2023), who finds that a large expansion of primary schooling in India raised wages for unskilled workers.

My empirical results contrast with Bedard (2001). While she finds that expanding higher education in the United States increased high school dropout rates, I find that students exposed to new HEIs in Brazil responded by increasing their high school persistence and completion rates. This difference likely reflects differences in context, and both results are consistent with my framework. First, average years of schooling in the United States in the 1970s were approximately 12 years, equivalent to high school completion, whereas average attainment in my setting is below 9 years, equivalent to primary school completion in Brazil. For an expansion in higher education to affect

high school completion, enough students must be dropping out of high school in the absence of college opportunities. This is true in my setting, but much less so in the United States. Second, in the signaling model of Bedard (2001), in the absence of HEIs, both low-type and high-type students graduate from high school in a pooling equilibrium. However, after an HEI opens, only high-type students attend college, while low-type students can, at most, complete high school. This leads to a separating equilibrium, allowing employers to correctly identify students' types and pay low wages to low types (high school dropouts or graduates) and high wages to high types (college graduates). In my framework, this is equivalent to the policy lowering wages for high school graduates, which in turn reduces high school completion and increases dropout, as found by Bedard (2001).

In a context more closely related to this paper, I also build on previous work by Jagnani and Khanna (2020) and provide further external validity for their findings. Importantly, I find significant gains in persistence at lower levels of education following higher-education expansion in a setting where most new HEIs are newly created, not prestigious, and predominantly technically focused (trade schools). I show that not only four-year colleges, but also smaller two-year trade schools can have important trickle-down effects on high school completion. By developing a theoretical framework and conducting an empirical analysis of the mechanisms, I explore the channels through which these effects may operate and highlight the key role of skill premia and increased college access opportunities in shaping this relationship.

Overall, expanding higher education in Brazil shifted earlier educational decisions among pre-tertiary students in receiving municipalities, raising students' high school persistence and graduation rates. Students were also more likely to attend federal higher education institutions. This policy provides a relevant example of how the availability of future educational opportunities, coupled with an effective increase in access for low-income students, can raise the expected returns to basic education and generate meaningful improvements in human capital accumulation, particularly among disadvantaged students and in places where returns to high school alone are not enough to keep students in school.

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A Additional Figures and Tables

Table A1: Graduation Outcomes Adjusted by 12th-grade Dropout Rates

	9th grade baseline		10th grade baseline	
	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)
Treated	0.001	0.011**	0.000	0.010***
	(0.004)	(0.004)	(0.004)	(0.004)
Observations	19,208	14,410	21,616	16,814
Pre-policy Mean	0.40	0.49	0.46	0.54

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. Outcomes are the adjusted versions of the persistence measures. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Graduation Outcomes by Baseline Grade

	9th grade baseline		10th grade baseline	
	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)
Treated	0.000	0.011**	-0.003	0.008**
	(0.004)	(0.004)	(0.003)	(0.003)
Observations	19,208	14,410	21,616	16,814
Pre-policy Mean	0.44	0.54	0.46	0.54

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: High School Completion by Baseline Grade, by Type of HEI Received

	9th grade baseline		10th grade baseline	
	Graduate on time	Eventually graduate	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)
Received a Trade School	-0.002	0.010**	-0.004	0.010**
	(0.005)	(0.005)	(0.004)	(0.004)
Observations	18,616	13,966	20,950	16,296
Pre-policy Mean	0.45	0.54	0.47	0.55
Received a University	0.011	0.021*	-0.007	-0.003
	(0.010)	(0.012)	(0.008)	(0.010)
Observations	16,544	12,412	18,619	14,483
Pre-policy Mean	0.44	0.54	0.44	0.53
P-value (diff)	0.27	0.42	0.74	0.21
Bootstrap p-value (diff)	0.26	0.40	0.77	0.23

Note: Standard errors in parentheses, clustered at the municipality level. Each type of HEI is compared with the same never-treated municipalities. Graduate on time is the share of students in the baseline-grade cohort who complete high school on time; Eventually graduate allows for up to a three-year delay. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). P-value (diff) tests the equality of the trade school and university estimates using $\sqrt{se_{TS}^2 + se_U^2}$, which ignores that both groups share the same comparison municipalities. Bootstrap p-value (diff) uses the standard error of the difference from a municipality-level bootstrap (200 replications, stratified by opening cohort), re-estimating both ATTs in each draw. Observations vary between outcomes due to years of data availability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Persistence in Secondary School (No Controls)

		Dropout rate					
	Start HS	9th grade	10th grade	11th grade	12th grade	Graduate on time	Eventually graduate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated	0.003	-0.110	-1.340***	-1.068***	-0.925***	-0.003	0.005
	(0.003)	(0.137)	(0.286)	(0.236)	(0.202)	(0.003)	(0.003)
Observations	19,208	35,241	35,241	35,241	35,241	21,616	16,814
Pre-policy Mean	0.79	7.57	18.21	14.31	10.62	0.46	0.54

Note: Standard errors in parentheses, clustered at the municipality level. No municipality-level control variables are included: ATTs are estimated under unconditional parallel trends and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. Dropout estimates (columns 2–5) use a common sample of municipality-years with non-missing dropout rates for all four grades. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Performance Outcomes (No Controls)

	9th Grade Math	9th Grade Portuguese	College Exam Scores	Share Taking Exam
	(1)	(2)	(3)	(4)
Treated	0.01	0.01	-0.51	0.03***
	(0.01)	(0.01)	(1.26)	(0.00)
Observations	17,548	17,548	16,485	16,485
Pre-policy Mean	-0.17	-0.39	525.48	0.10

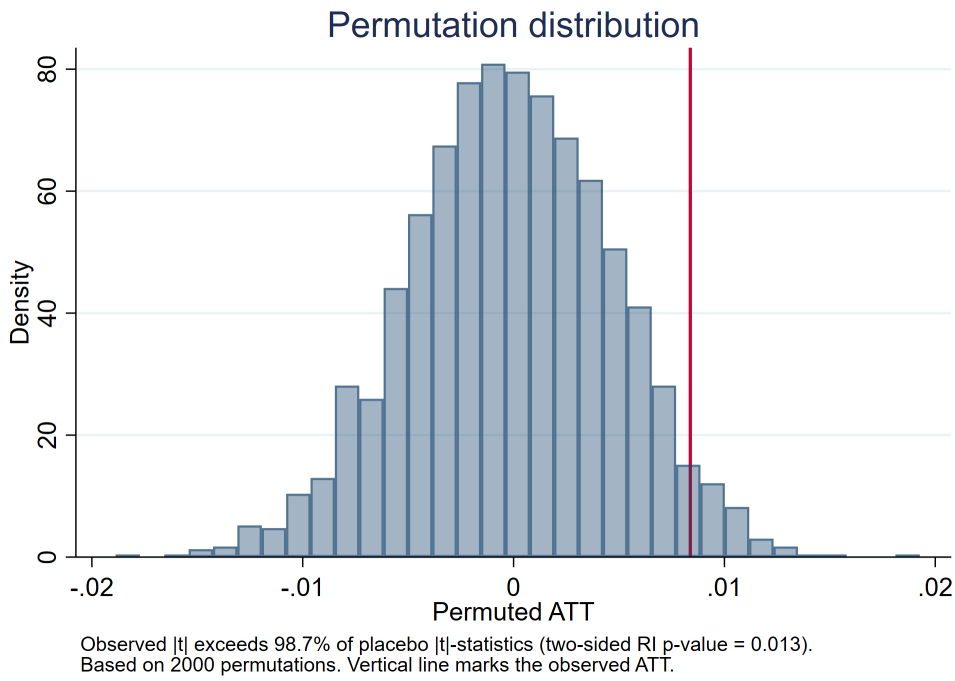
Note: Standard errors in parentheses, clustered at the municipality level. No municipality-level control variables are included: ATTs are estimated under unconditional parallel trends and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Robustness: Excluding Students in Technical High Schools

	Graduate on time		Eventually graduate	
	All students	Excl. technical HS	All students	Excl. technical HS
	(1)	(2)	(3)	(4)
Treated	-0.003	-0.005	0.008**	0.005
	(0.003)	(0.004)	(0.003)	(0.004)
p-value: equal to all students		0.789		0.509
Observations	21,616	21,616	16,814	16,814
Pre-policy Mean	0.46	0.46	0.54	0.54

Note: 10th-grade baseline. Columns (2) and (4) exclude students who attended technical high schools. Standard errors in parentheses, clustered at the municipality level. All specifications include pre-policy municipality-level controls, discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). The p-value tests whether the estimate excluding technical high school students equals the full-sample estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Permutation Test: Distribution of Placebo Estimates



Note: Distribution of ATT estimates from 2,000 random reassignments of treatment cohorts (including never-treated status) across municipalities, holding fixed the number of municipalities in each cohort. The vertical line marks the observed estimate. Outcome: eventually graduating from high school, 10th grade baseline.

Table A7: Effects of Higher Education Expansion on Student Composition

Panel A: *5th Grade*	White	Studied in Public Schools Only	Mother Completed High School
	(1)	(2)	(3)
Treated	-0.008*	-0.012	0.000
	(0.004)	(0.007)	(0.004)
Observations	14,470	14,471	14,471
Pre-policy Mean	0.35	0.57	0.29
Panel B: *9th Grade*	White	Studied in Public Schools Only	Mother Completed High School
	(1)	(2)	(3)
Treated	0.002	0.020***	-0.001
	(0.003)	(0.005)	(0.004)
Observations	15,093	15,093	15,092
Pre-policy Mean	0.35	0.90	0.31

Note: Standard errors in parentheses, clustered at the municipality level. All specifications include municipality-level control variables, measured pre-policy and discretized. ATTs are calculated using doubly robust inverse probability weighting (DR-IPW) and aggregated to generate an overall average effect following Callaway and Sant'Anna (2021). Observations vary between outcomes due to years of data availability. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

B Complete overview of the model

B.1 Setup

At the beginning of 10th grade, the first grade of high school in Brazil, a student enrolled in high school chooses a schooling path $(s, d) \in \{(0,0), (1,0), (1,1)\}$, where $s = 1$ denotes completing high school, $s = 0$ denotes dropping out of high school before completion, and $d = 1$ denotes attempting college. Because college requires first graduating high school, the pair $(0, 1)$ is not feasible. The student maximizes expected utility:

$$\max_{(s,d)\in\{(0,0),(1,0),(1,1)\}} U(C_1) + \beta_2 \,\mathbb{E}\big[U(C_2)\big] + \beta_3 \,\mathbb{E}\big[U(C_3)\big], \tag{B1}$$

where $U(c) = \ln c$ and the expectation is over the college admission lottery. For paths $(0, 0)$ and $(1, 0)$ there is no uncertainty, so the expectation is trivial. For path $(1, 1)$, the student is admitted with probability $\pi_c \in (0, 1)$; conditional on admission they enroll and pay costs p^c, and conditional on non-admission they earn high-school wages. When choosing s and d, a student knows their cost

of high school attendance, p^H, college cost, p^c, and the likelihood of being accepted to an HEI, π_c, but does not know the college admission outcome.

Period 1. Period 1 lasts 3 years (the duration of high school in Brazil, grades 10–12):

$$C_1 = \bar{k}_1 + (1-s)\, y_1^{NH} - s\, p^H,$$

where $\bar{k}_1 = 3\bar{k}$ and $y_1^{NH} = 3y^{NH}$.

Period 2. Period 2 lasts 4 years (the typical higher education duration), with $\bar{k}_2 = 4\bar{k}$:

$$C_2 = \begin{cases} \bar{k}_2 + y_2^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k}_2 + y_2^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted (probability } 1-\pi_c), \\ \bar{k}_2 - p^c & \text{if } (s,d) = (1,1) \text{ and admitted (probability } \pi_c), \end{cases}$$

where $y_2^{NH} = 4y^{NH}$ and $y_2^{HS} = 4y^{HS}$.

Period 3. Period 3 represents the remainder of working life. C_3 denotes *per-year* consumption in that period, which is assumed to be constant:

$$C_3 = \begin{cases} \bar{k} + y^{NH} & \text{if } (s,d) = (0,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,0), \\ \bar{k} + y^{HS} & \text{if } (s,d) = (1,1) \text{ and not admitted}, \\ \bar{k} + y^{C} & \text{if } (s,d) = (1,1) \text{ and admitted}. \end{cases}$$

Timing and discount factors. Period 1 lasts 3 years, period 2 lasts 4 years, and period 3 runs from year 8 to the end of working life, year N. Accordingly,

$$\beta_2 = \beta^3, \qquad \beta_3 \;\equiv\; \sum_{t=8}^{N} \beta^{t-1} \;=\; \beta^7\, \frac{1-\beta^{N-7}}{1-\beta} \;>\; 0. \tag{B2}$$

Because C_3 is measured per year, β_3 is the exact present value, as of period 1, of one unit of per-year consumption received in every year $t = 8, \ldots, N$.

Wages. y^{NH} is the wage for a worker who did not complete high school, y^{HS} for a high-school graduate, and y^{C} for a college graduate. I assume all are constant over time. An individual's education level is fixed after the schooling decision is made and the college admission lottery is realized.

Endowment and costs. $\bar{k}$ is the household endowment per year. $p^H > 0$ is the direct cost of remaining enrolled in and completing high school and $p^c > 0$ is the direct cost of attending college. Both p^H and p^c do not include tuition costs, and are mostly comprised of transportation or relocation costs. Throughout, I assume $p^H < \bar{k}_1$ and $p^c < \bar{k}_2$ so that period-1 and period-2 consumption are strictly positive on all paths.

College access. $\pi_c \in (0,1)$ is the probability of admission to a higher-education institution. The student knows π_c when choosing (s,d), but the admission outcome is only realized after the choice.

Cost distributions. Costs are drawn independently and uniformly: $p^H \sim U[0, \bar{P}^H]$ and $p^c \sim U[0, \bar{P}^c]$.

B.2 Value Functions

Substituting $U(c) = \ln c$ and taking the expectation over the admission lottery for path $(1,1)$:

$$V(0,0) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}), \tag{B3}$$

$$V(1,0) = \ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}), \tag{B4}$$

$$\begin{aligned} V(1,1) = \ln(\bar{k}_1 - p^H) &+ \beta_2 \big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] \\ &+ \beta_3 \big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^{C})\big]. \end{aligned} \tag{B5}$$

B.3 Decision Thresholds and Decision Rule

B.3.1 Threshold 1: $(1,0)$ vs. $(0,0)$

Derivation. A student chooses $(1,0)$ over $(0,0)$ when $V(1,0) \geq V(0,0)$. Setting $V(1,0) = V(0,0)$ using (B4) and (B3):

$$\ln(\bar{k}_1 - p^H) + \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}) = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}).$$

Isolating p^H on one side, I can find a threshold for which a student will be indifferent between $(1,0)$ and $(0,0)$:

$$\begin{aligned}
&\ln(\bar{k}_1 - p^H) - \ln(\bar{k}_1 + y_1^{NH}) = \beta_2\big[\ln(\bar{k}_2 + y_2^{NH}) - \ln(\bar{k}_2 + y_2^{HS})\big] + \beta_3\big[\ln(\bar{k} + y^{NH}) - \ln(\bar{k} + y^{HS})\big] \\
\implies \quad &\ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) = \beta_2 \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right) \\
\implies \quad &\ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) = \ln\left[\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2}\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}\right] \\
\implies \quad &\bar{p}_{10}^H = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2}\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}.
\end{aligned}$$

Since $V(1,0) - V(0,0)$ is strictly decreasing in p^H, a student chooses $(1,0)$ over $(0,0)$ if and only if $p^H \leq \bar{p}_{10}^H$.

B.3.2 Threshold 2: $(1,1)$ vs. $(1,0)$

Derivation. A student chooses $(1,1)$ over $(1,0)$ when $V(1,1) \geq V(1,0)$. Setting $V(1,1) = V(1,0)$. The terms $\ln(\bar{k}_1 - p^H)$ are identical in (B5) and (B4) and cancel, leaving

$$\begin{aligned}
&\beta_2\big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] + \beta_3\big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^C)\big] \\
&\quad = \beta_2 \ln(\bar{k}_2 + y_2^{HS}) + \beta_3 \ln(\bar{k} + y^{HS}).
\end{aligned}$$

Subtract the right side from the left. In the β_2 block, $(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) - \ln(\bar{k}_2 + y_2^{HS}) = -\pi_c \ln(\bar{k}_2 + y_2^{HS})$, and similarly in the β_3 block:

$$\pi_c\beta_2\big[\ln(\bar{k}_2 - p^c) - \ln(\bar{k}_2 + y_2^{HS})\big] + \pi_c\beta_3\big[\ln(\bar{k} + y^C) - \ln(\bar{k} + y^{HS})\big] = 0.$$

Dividing through by $\pi_c > 0$ and $\beta_2 > 0$, and exponentiating:

$$\frac{\bar{k}_2 - p^c}{\bar{k}_2 + y_2^{HS}} = \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}.$$

Multiplying by $(\bar{k}_2 + y_2^{HS}) > 0$ and subtracting from $\bar{k}_2$:

$$\bar{p}_{11|10}^c = \bar{k}_2 - (\bar{k}_2 + y_2^{HS})\left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2}. \tag{B6}$$

Since $V(1,1) - V(1,0)$ is strictly decreasing in p^c, a student chooses $(1,1)$ over $(1,0)$ if and only if $p^c \leq \bar{p}^c_{11|10}$. Additionally, since $(1,0)$ is available for that student, it must be true that $p^H \leq \bar{p}^H_{10}$.

B.3.3 Threshold 3: $(1,1)$ vs. $(0,0)$

Derivation. A student chooses $(1,1)$ over $(0,0)$ when $V(1,1) \geq V(0,0)$. Setting $V(1,1) = V(0,0)$:

$$\begin{aligned}\ln(\bar{k}_1 - p^H) + \beta_2\big[(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) + \pi_c \ln(\bar{k}_2 - p^c)\big] + \beta_3\big[(1-\pi_c)\ln(\bar{k} + y^{HS}) + \pi_c \ln(\bar{k} + y^C)\big] \\ = \ln(\bar{k}_1 + y_1^{NH}) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) + \beta_3 \ln(\bar{k} + y^{NH}).\end{aligned}$$

Isolating p^c:

$$\begin{aligned}\pi_c\beta_2 \ln(\bar{k}_2 - p^c) = \ln\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right) + \beta_2 \ln(\bar{k}_2 + y_2^{NH}) - \beta_2(1-\pi_c)\ln(\bar{k}_2 + y_2^{HS}) \\ + \beta_3 \ln(\bar{k} + y^{NH}) - \beta_3(1-\pi_c)\ln(\bar{k} + y^{HS}) - \beta_3\pi_c \ln(\bar{k} + y^C).\end{aligned}$$

Divide every term by $\pi_c\beta_2 > 0$ and rearranging:

$$\ln(\bar{k}_2 - p^c) = \frac{1}{\pi_c}\left\{\frac{1}{\beta_2}\ln\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right) + \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}}\right) + \frac{\beta_3}{\beta_2}\ln\left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)\right\}.$$

$$\bar{p}^c_{11|00}(p^H) = \bar{k}_2 - \big[E(p^H)\big]^{1/\pi_c}, \tag{B7}$$

where, written out in full,

$$E(p^H) \;\equiv\; \left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - p^H}\right)^{1/\beta_2} \cdot \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \cdot \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)^{\beta_3/\beta_2} \;>\; 0. \tag{B8}$$

By the same monotonicity argument as in Section B.3.2, a student chooses $(1,1)$ over $(0,0)$ if and only if $p^c \leq \bar{p}^c_{11|00}(p^H)$. Since a student is comparing $(1,1)$ to $(0,0)$ and not $(1,0)$, it must be true that $p^H > \bar{p}^H_{10}$ and $(1,0)$ is not available for that student. Note that $\bar{p}^c_{11|00}$ is a *function* of p^H, since p^H appears in $E(p^H)$.

Boundary condition $\bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10}$. Evaluating $\bar{p}^c_{11|00}$ at $p^H = \bar{p}^H_{10}$ and using

$$\Pi \;\equiv\; \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2}\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3} \;>\; 0, \qquad \text{so} \qquad \bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\,\Pi.$$

we have that $\bar{k}_1 - \bar{p}_{10}^H = (\bar{k}_1 + y_1^{NH})\Pi$, so:

$$\left(\frac{\bar{k}_1 + y_1^{NH}}{\bar{k}_1 - \bar{p}_{10}^H}\right)^{1/\beta_2} = \Pi^{-1/\beta_2} = \frac{\bar{k}_2 + y_2^{HS}}{\bar{k}_2 + y_2^{NH}} \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^{NH}}\right)^{\beta_3/\beta_2}.$$

Substituting into (B8), the period-2 factors give

$$\frac{\bar{k}_2 + y_2^{HS}}{\bar{k}_2 + y_2^{NH}} \cdot \frac{\bar{k}_2 + y_2^{NH}}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} = (\bar{k}_2 + y_2^{HS})^{\pi_c},$$

since $(\bar{k}_2 + y_2^{NH})$ cancels and $x/x^{1-\pi_c} = x^{\pi_c}$; and the period-3 factors give

$$\left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^{NH}} \cdot \frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)^{\beta_3/\beta_2} = \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\pi_c\beta_3/\beta_2}.$$

Hence $E(\bar{p}_{10}^H) = (\bar{k}_2 + y_2^{HS})^{\pi_c}\big((\bar{k} + y^{HS})/(\bar{k} + y^C)\big)^{\pi_c\beta_3/\beta_2}$, and raising to $1/\pi_c$ divides every exponent by π_c:

$$\bar{k}_2 - \bar{p}_{11|00}^c(\bar{p}_{10}^H) = (\bar{k}_2 + y_2^{HS}) \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C}\right)^{\beta_3/\beta_2} = \bar{k}_2 - \bar{p}_{11|10}^c.$$

Therefore

$$\bar{p}_{11|00}^c(\bar{p}_{10}^H) = \bar{p}_{11|10}^c. \tag{B9}$$

This is the internal consistency condition of the model: at $p^H = \bar{p}_{10}^H$ the student is indifferent between $(1, 0)$ and $(0, 0)$, so comparing $(1, 1)$ against either must yield the same critical college cost.

Strict monotonicity of $\bar{p}_{11|00}^c(\cdot)$. Writing $\big((\bar{k}_1 + y_1^{NH})/(\bar{k}_1 - p^H)\big)^{1/\beta_2} = (\bar{k}_1 + y_1^{NH})^{1/\beta_2}(\bar{k}_1 - p^H)^{-1/\beta_2}$ and raising $E(p^H)$ to $1/\pi_c$, the p^H-dependent factor becomes $(\bar{k}_1 - p^H)^{-1/(\pi_c\beta_2)}$. Setting $\alpha \equiv 1/(\pi_c\beta_2) > 0$,

$$\bar{p}_{11|00}^c(p^H) = \bar{k}_2 - \Lambda\,(\bar{k}_1 - p^H)^{-\alpha}, \Lambda \equiv \left[\frac{(\bar{k}_1 + y_1^{NH})^{1/\beta_2}(\bar{k}_2 + y_2^{NH})}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)^{\beta_3/\beta_2}\right]^{1/\pi_c} > 0, \tag{B10}$$

where Λ contains no p^H. Differentiating:

$$\frac{\partial \bar{p}_{11|00}^c(p^H)}{\partial p^H} = -\Lambda\,\alpha\,(\bar{k}_1 - p^H)^{-\alpha-1} < 0.$$

So $\bar{p}^c_{11|00}(\cdot)$ is strictly decreasing, and with (B9),

$$\bar{p}^c_{11|00}(p^H) < \bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10} \qquad \text{for every } p^H > \bar{p}^H_{10}. \tag{B11}$$

A student who has paid a larger p^H has lower period-1 consumption, so the critical college cost at which she is willing to take the college path is lower. A useful identity, immediate from (B10), is

$$\Lambda\,(\bar{k}_1 - p^H)^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|00}(p^H), \qquad \text{and in particular} \qquad \Lambda\,(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|10}. \tag{B12}$$

B.3.4 Decision Rule

Combining the three thresholds:

$$(s,d)^* = \begin{cases} (1,1) & \text{if } \left[p^H \le \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|10}\right] \text{ or } \left[p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\right], \\ (1,0) & \text{if } p^H \le \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|10}, \\ (0,0) & \text{if } p^H > \bar{p}^H_{10} \text{ and } p^c > \bar{p}^c_{11|00}(p^H). \end{cases} \tag{B13}$$

The rule is exhaustive and the four cases are internally consistent:

- $p^H \le \bar{p}^H_{10}$, $p^c \le \bar{p}^c_{11|10}$: Threshold 1 gives $(1,0) \succ (0,0)$ and Threshold 2 gives $(1,1) \succ (1,0)$, so by transitivity $(1,1)$ is chosen.
- $p^H \le \bar{p}^H_{10}$, $p^c > \bar{p}^c_{11|10}$: $(1,0) \succ (0,0)$ and $(1,0) \succ (1,1)$, so $(1,0)$ is chosen.
- $p^H > \bar{p}^H_{10}$, $p^c \le \bar{p}^c_{11|00}(p^H)$: $(0,0) \succ (1,0)$, so $(1,0)$ is dominated; Threshold 3 gives $(1,1) \succ (0,0)$ and transitivity gives $(1,1) \succ (1,0)$, so $(1,1)$ is chosen.
- $p^H > \bar{p}^H_{10}$, $p^c > \bar{p}^c_{11|00}(p^H)$: $(0,0)$ dominates both alternatives.

B.4 Aggregate High School Graduation

Definition. S^* is the aggregate share of students who graduate high school, i.e. who choose $s = 1$ either through $(1,0)$ or $(1,1)$.

Derivation. From (B13), a student graduates in exactly one of two mutually exclusive ways: either $p^H \le \bar{p}^H_{10}$, in which case she chooses $(1,0)$ or $(1,1)$ depending on p^c but graduates either way,

or $p^H > \bar{p}^H_{10}$ and $p^c \le \bar{p}^c_{11|00}(p^H)$, in which case she moves directly from $(0,0)$ to $(1,1)$. Hence

$$S^* = \Pr(p^H \le \bar{p}^H_{10}) + \Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\big).$$

Since the threshold $\bar{p}^c_{11|00}(p^H)$ depends on the realization of p^H, I apply the law of total probability and conditioning on p^H to get:

$$\Pr\big(p^H > \bar{p}^H_{10} \text{ and } p^c \le \bar{p}^c_{11|00}(p^H)\big) = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c \le \bar{p}^c_{11|00}(p^H) \,\big|\, p^H\big)\, f(p^H)\, dp^H,$$

with $f(p^H) = 1/\bar{P}^H$. Because the two cost draws are *independent random variables*, the conditional probability equals the unconditional distribution function of p^c evaluated at the realized threshold: $\Pr(p^c \le \bar{p}^c_{11|00}(p^H) \mid p^H) = \bar{p}^c_{11|00}(p^H)/\bar{P}^c$, valid whenever $\bar{p}^c_{11|00}(p^H) \in [0, \bar{P}^c]$, which I assume throughout (Assumption 2 below), together with $\bar{p}^H_{10} \in (0, \bar{P}^H)$. Writing $\theta_H \equiv \bar{p}^H_{10}/\bar{P}^H$,

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{B14}$$

[Regularity] $\bar{p}^H_{10} \in (0, \bar{P}^H)$.

[Interior solution] $\bar{p}^c_{11|00}(p^H) \in (0, \bar{P}^c)$ for all $p^H \in [\bar{p}^H_{10}, \bar{P}^H]$.

Closed form. Substituting $\bar{p}^c_{11|00}(p^H)$ below into (B14)

$$\bar{p}^c_{11|00}(p^H) = \bar{k}_2 - \Lambda\,(\bar{k}_1 - p^H)^{-\alpha}, \Lambda \equiv \left[\frac{(\bar{k}_1 + y_1^{NH})^{1/\beta_2}(\bar{k}_2 + y_2^{NH})}{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}} \left(\frac{\bar{k} + y^{NH}}{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}\right)^{\beta_3/\beta_2}\right]^{1/\pi_c} > 0,$$

We have:

$$\int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H = \bar{k}_2(\bar{P}^H - \bar{p}^H_{10}) - \Lambda \int_{\bar{p}^H_{10}}^{\bar{P}^H} (\bar{k}_1 - p^H)^{-\alpha}\, dp^H,$$

and substituting $u = \bar{k}_1 - p^H$, $du = -dp^H$,

$$\int_{\bar{p}^H_{10}}^{\bar{P}^H} (\bar{k}_1 - p^H)^{-\alpha}\, dp^H = \int_{\bar{k}_1 - \bar{P}^H}^{\bar{k}_1 - \bar{p}^H_{10}} u^{-\alpha}\, du = \frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha},$$

valid for $\alpha \neq 1$, i.e. $\pi_c \beta_2 \neq 1$. The expression is strictly positive in both cases: for $\alpha < 1$ the numerator is positive and $1-\alpha > 0$; for $\alpha > 1$ both are negative. Hence

$$S^* = \theta_H + \frac{1}{\bar{P}^H \bar{P}^c} \underbrace{\left\{ \bar{k}_2(\bar{P}^H - \bar{p}^H_{10}) - \Lambda \, \frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha} \right\}}_{\equiv \; \mathcal{S} \; = \; \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H} \tag{B15}$$

B.4.1 Monotonicity of S^*

If the share of individuals completing high school as represented by S^* is a monotonic function of the thresholds, I can analyze changes in the thresholds to understand how policy-induced changes in parameters can affect S^*.

Independence from $\bar{p}^c_{11|10}$. $\bar{p}^c_{11|10}$ governs only the split between $(1,0)$ and $(1,1)$ among students with $p^H \leq \bar{p}^H_{10}$. Both choices involve graduating, so moving a student between them leaves S^* unchanged. Therefore, $\bar{p}^c_{11|10}$ appears in neither (B14) nor (B15).

S^* is strictly increasing in $\bar{p}^H_{10}$. Differentiating (B15) directly: $\partial \theta_H / \partial \bar{p}^H_{10} = 1/\bar{P}^H$ and $\partial \big[\bar{k}_2(\bar{P}^H - \bar{p}^H_{10})\big] / \partial \bar{p}^H_{10} = -\bar{k}_2$. Since only $(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha}$ depends on $\bar{p}^H_{10}$ in the remaining term, and by the chain rule, we have that $\partial (\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} / \partial \bar{p}^H_{10} = -(1-\alpha)(\bar{k}_1 - \bar{p}^H_{10})^{-\alpha}$, then:

$$\frac{\partial}{\partial \bar{p}^H_{10}} \left[-\Lambda \, \frac{(\bar{k}_1 - \bar{p}^H_{10})^{1-\alpha} - (\bar{k}_1 - \bar{P}^H)^{1-\alpha}}{1-\alpha} \right] = +\Lambda \, (\bar{k}_1 - \bar{p}^H_{10})^{-\alpha},$$

By (B12), we have that $\Lambda \, (\bar{k}_1 - \bar{p}^H_{10})^{-\alpha} = \bar{k}_2 - \bar{p}^c_{11|10}$. Replacing it and combining both derivatives:

$$\frac{\partial S^*}{\partial \bar{p}^H_{10}} = \frac{1}{\bar{P}^H} - \frac{\bar{p}^c_{11|10}}{\bar{P}^H \bar{P}^c} = \frac{1}{\bar{P}^H} \left(1 - \frac{\bar{p}^c_{11|10}}{\bar{P}^c} \right) = \frac{1 - \theta_{c,10}}{\bar{P}^H} > 0, \tag{B16}$$

with $\theta_{c,10} \equiv \bar{p}^c_{11|10} / \bar{P}^c$, strictly positive because $\theta_{c,10} < 1$ under Assumption B.4.

S^* is strictly increasing in the college threshold function. Using the integral form of (B14), in which $\bar{p}^c_{11|00}(\cdot)$ appears explicitly as the integrand. Since S^* is an affine functional of $\bar{p}^c_{11|00}(\cdot)$ with strictly positive coefficient $1/(\bar{P}^H \bar{P}^c)$, the functional derivative at any $p^H_0 \in (\bar{p}^H_{10}, \bar{P}^H)$ is

$$\frac{\delta S^*}{\delta \, \bar{p}^c_{11|00}(p^H_0)} = \frac{1}{\bar{P}^H \bar{P}^c} > 0.$$

So any parameter change that raises $\bar{p}^c_{11|00}(p^H)$ within $[\bar{p}^H_{10}, \bar{P}^H]$, and lowers it nowhere, strictly raises S^*.

B.5 Necessary Conditions

The share of individuals completing high school might move as a result of changes in two channels: the indirect labor market channels, which moves students from $(0,0)$ to $(1,0)$ and increases S^* by increasing θ_H , and the direct education effect via transitions from $(0,0)$ to $(1,1)$, which operates through the second term. In this section, I derive the necessary conditions for these transitions to take place.

B.5.1 Condition 1: a non-empty pool at $(0,0)$, $\bar{p}^H_{10} < \bar{P}^H$

For either channel to be active, the path $(0,0)$ need to attract at least some students pre-policy. A student is at $(0,0)$ if and only if $p^H > \bar{p}^H_{10}$ *and* $p^c > \bar{p}^c_{11|00}(p^H)$. Conditioning on p^H and using independence of the two draws,

$$\begin{aligned}\Pr\big((s,d)^* = (0,0)\big) &= \int_{\bar{p}^H_{10}}^{\bar{P}^H} \Pr\big(p^c > \bar{p}^c_{11|00}(p^H) \,\big|\, p^H\big)\, f(p^H)\, dp^H = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\bar{P}^c - \bar{p}^c_{11|00}(p^H)}{\bar{P}^c} \cdot \frac{1}{\bar{P}^H}\, dp^H \\ &= \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \big[\bar{P}^c - \bar{p}^c_{11|00}(p^H)\big]\, dp^H,\end{aligned}$$

where the middle step uses $\Pr(p^c > \bar{p}^c_{11|00}(p^H)) = 1 - \bar{p}^c_{11|00}(p^H)/\bar{P}^c$, which is valid because Assumption B.4 places $\bar{p}^c_{11|00}(p^H)$ inside $(0, \bar{P}^c)$.

This condition implies that $\bar{p}^H_{10} < \bar{P}^H$. Assume $\bar{p}^H_{10} \geq \bar{P}^H$. Then every student draws $p^H \leq \bar{P}^H \leq \bar{p}^H_{10}$, so every student graduates and $S^* = 1$. Hence no parameter change can raise S^*, and $\bar{p}^H_{10} < \bar{P}^H$ is necessary. Recall the formula for $\bar{p}^H_{10}$:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right)^{\beta_3}.$$

As $y^{HS} \to \infty$ both ratios tend to 0 and $\bar{p}^H_{10} \to \bar{k}_1$. If $\bar{k}_1 \geq \bar{P}^H$, then $\theta_H \to 1$ and the pool at $(0,0)$ disappears. Condition 1 therefore places an *upper* bound on the high-school wage premium: it must not be so large that all students graduate even in the complete absence of any college option.

B.5.2 Condition 2: a reachable pool

Condition 1 guarantees that students sit at $(0,0)$. However, it does not guarantee that either transitions can take place. That is, that students sitting at $(0,0)$ can move to either $(1,0)$ or $(1,1)$. Below, I derive the necessary conditions for each transition to take place.

Necessary conditions for the indirect labor market channel. For $(0,0)$ to $(1,0)$ transitions to take place, it needs to be true that $p^H < \bar{p}^H_{10}$ is attainable for at least some students. That implies that $\bar{p}^H_{10}$ must be positive, since $p^H > 0$.

From:

$$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}} \right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}} \right)^{\beta_3}.$$

$\bar{p}^H_{10} > 0$ if and only if $(\bar{k}_1 + y_1^{NH}) \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}} \right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}} \right)^{\beta_3} < \bar{k}_1$. Dividing by $(\bar{k}_1 + y_1^{NH}) > 0$ and substituting $\bar{k}_1 = 3\bar{k}$, $y_1^{NH} = 3y^{NH}$ on the right,

$$\frac{\bar{k}_1}{\bar{k}_1 + y_1^{NH}} = \frac{3\bar{k}}{3(\bar{k} + y^{NH})} = \frac{\bar{k}}{\bar{k} + y^{NH}},$$

so the exact necessary and sufficient condition is

$$\bar{p}^H_{10} > 0 \iff \left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}} \right)^{\beta_2} \left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}} \right)^{\beta_3} < \frac{\bar{k}}{\bar{k} + y^{NH}}. \tag{B17}$$

If $y^{HS} \leq y^{NH}$ then $y_2^{HS} \leq y_2^{NH}$, so both ratios are ≥ 1. Since $y^{NH} > 0$, $\bar{k}/(\bar{k} + y^{NH}) < 1$. Thus, $y^{HS} \leq y^{NH}$ implies that $\bar{p}^H_{10} \leq 0$. Contrapositively, $\bar{p}^H_{10} > 0 \Rightarrow y^{HS} > y^{NH}$.

Necessary conditions for the direct education channel. For $(0,0)$ to $(1,1)$ transitions to take place, it needs to be true that $p^c < \bar{p}^c_{11|00}$ is attainable for at least some students. That implies that $\bar{p}^c_{11|00}$ is positive, since $p^c > 0$. By (B11), if $\bar{p}^c_{11|10} \leq 0$, then $\bar{p}^c_{11|00} < 0$. Thus, $\bar{p}^c_{11|10} > 0$ implies $\bar{p}^c_{11|00} > 0$. I can then study when $\bar{p}^c_{11|10} > 0$ to determine the necessary conditions for $\bar{p}^c_{11|00} > 0$.

Using $\bar{k}_2 = 4\bar{k}$ and $\bar{k}_2 + y_2^{HS} = 4(\bar{k} + y^{HS})$:

$$\bar{p}^c_{11|10} = 4 \left[\bar{k} - (\bar{k} + y^{HS}) \left(\frac{\bar{k} + y^{HS}}{\bar{k} + y^C} \right)^{\beta_3/\beta_2} \right].$$

Note that at $y^C = y^{HS}$ the ratio equals 1 and $\bar{p}^c_{11|10} = 4[\bar{k} - (\bar{k} + y^{HS})] = -4y^{HS} < 0$. As $y^C \to \infty$ the ratio tends to 0 and $\bar{p}^c_{11|10} \to 4\bar{k} > 0$. Since $\bar{p}^c_{11|10}$ is increasing in y^C, there is a

unique threshold $\bar{y}^C$ where $\bar{p}^c_{11|10} = 0$, and for which any $y^C > \bar{y}^C$ implies that $\bar{p}^c_{11|10} > 0$. Then $\bar{p}^c_{11|10} > 0 \iff y^C > \bar{y}^C$. Setting $\bar{p}^c_{11|10} = 0$ and dividing by $4(\bar{k} + y^{HS})$:

$$\frac{\bar{k}}{\bar{k} + y^{HS}} = \left(\frac{\bar{k} + y^{HS}}{\bar{k} + \bar{y}^C}\right)^{\beta_3/\beta_2}.$$

Raising both sides to the power $\beta_2/\beta_3 > 0$ and multiplying by $(\bar{k} + y^{HS})$:

$$\bar{y}^C = (\bar{k} + y^{HS})\left(\frac{\bar{k} + y^{HS}}{\bar{k}}\right)^{\beta_2/\beta_3} - \bar{k}. \tag{B18}$$

Since $y^{HS} > 0$, it follows that $\left((\bar{k} + y^{HS})/\bar{k}\right)^{\beta_2/\beta_3} > 1$, and thus $\bar{y}^C > y^{HS}$ strictly. Therefore, a necessary condition for $\bar{p}^c_{11|10} > 0$, and consequently, $\bar{p}^c_{11|00} > 0$, is that $y^C > \bar{y}^C > y^{HS}$.

B.6 Comparative Statics

B.6.1 The chain rule

To analyze how the share of students completing high school changes with policy-driven changes in parameters, I differentiate the closed form expression (B15) with respect to key parameters. Recall $\mathcal{S}$ denotes the braced expression, and by construction $\mathcal{S} = \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H$. For a given parameter x,

$$\frac{\partial S^*}{\partial x} = \frac{1}{\bar{P}^H}\frac{\partial \bar{p}^H_{10}}{\partial x} + \frac{1}{\bar{P}^H \bar{P}^c}\frac{d\mathcal{S}}{dx}, \qquad \frac{d\mathcal{S}}{dx} = \frac{\partial \mathcal{S}}{\partial \bar{p}^H_{10}}\frac{\partial \bar{p}^H_{10}}{\partial x} + \left.\frac{\partial \mathcal{S}}{\partial x}\right|_{\bar{p}^H_{10} \text{ fixed}},$$

where $\partial \mathcal{S}/\partial \bar{p}^H_{10} = -\bar{p}^c_{11|10}$. Collecting the terms in $\partial \bar{p}^H_{10}/\partial x$ and using $1/\bar{P}^H - \bar{p}^c_{11|10}/(\bar{P}^H \bar{P}^c) = (1 - \theta_{c,10})/\bar{P}^H$:

$$\frac{\partial S^*}{\partial x} = \underbrace{\frac{1 - \theta_{c,10}}{\bar{P}^H} \cdot \frac{\partial \bar{p}^H_{10}}{\partial x}}_{\text{(0,0) to (1,0)}} + \underbrace{\frac{1}{\bar{P}^H \bar{P}^c}\int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial x}\, dp^H}_{\text{(0,0) to (1,1)}}. \tag{B19}$$

B.6.2 Signing the college-threshold term

Define $\Delta(p^c, x; p^H) \equiv V(1,1) - V(0,0)$:

$$V(1,1) - V(0,0) = \ln\left(\frac{\bar{k}_1 - p^H}{\bar{k}_1 + y_1^{NH}}\right) + \beta_2 \ln\left(\frac{(\bar{k}_2 + y_2^{HS})^{1-\pi_c}(\bar{k}_2 - p^c)^{\pi_c}}{\bar{k}_2 + y_2^{NH}}\right) + \beta_3 \ln\left(\frac{(\bar{k} + y^{HS})^{1-\pi_c}(\bar{k} + y^C)^{\pi_c}}{\bar{k} + y^{NH}}\right), \tag{B20}$$

So that $\Delta = 0$ at $p^c = \bar{p}^c_{11|00}(p^H)$. The only p^c-dependent term is $\pi_c\beta_2 \ln(\bar{k}_2 - p^c)$, so

$$\frac{\partial \Delta}{\partial p^c} = -\frac{\pi_c \beta_2}{\bar{k}_2 - p^c} < 0,$$

which is strictly negative because $p^c < \bar{k}_2$. The implicit function theorem therefore gives

$$\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial x} = -\frac{\partial \Delta / \partial x}{\partial \Delta / \partial p^c} = \frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} \cdot \frac{\partial \Delta}{\partial x}, \tag{B21}$$

Since $\frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} > 0$, then $\operatorname{sign}\big(\partial \bar{p}^c_{11|00}/\partial x\big) = \operatorname{sign}\big(\partial \Delta / \partial x\big)$, and it suffices to sign $\partial \Delta / \partial x$.

B.6.3 Effect of p^c via the support $\bar{P}^c$

$\bar{P}^c$ enters S^* only through the explicit factor $1/\bar{P}^c$ in (B14), and

$$\frac{\partial S^*}{\partial \bar{P}^c} = -\frac{1}{\bar{P}^H \, \bar{P}^{c2}} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \bar{p}^c_{11|00}(p^H)\, dp^H. \tag{B22}$$

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, $\bar{p}^c_{11|00}(\bar{p}^H_{10}) = \bar{p}^c_{11|10} > 0$ by (B9) and, by continuity, $\bar{p}^c_{11|00} > 0$ on a set of positive measure, so the integral is strictly positive and (B22) is strictly negative. On the other hand, if the necessary condition is not met and If $\bar{p}^c_{11|10} \leq 0$, then by (B11) $\bar{p}^c_{11|00}(p^H) \leq 0$ throughout the range, and $\Pr(p^c \leq \bar{p}^c_{11|00}(p^H)) = 0$ for every student. The integral is zero, and $\partial S^*/\partial \bar{P}^c = 0$. Hence

$$\frac{\partial S^*}{\partial \bar{P}^c} < 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0, \tag{B23}$$

so lowering college costs unambiguously raises S^* exactly when the college channel is non-empty.

B.6.4 Effect of π_c

Changes in S^* brought about by changes in π_c run only through the educational opportunities channel, so $\partial \bar{p}^H_{10}/\partial \pi_c = 0$. Then, through the education opportunities channel, we have that:

$$\frac{\partial \Delta}{\partial \pi_c} = \beta_2 \ln\left(\frac{\bar{k}_2 - p^c}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^C}{\bar{k} + y^{HS}}\right), \tag{B24}$$

The first term of (B24) is strictly negative for *every* student and every admissible p^c, because $y_2^{HS} > 0$ and $p^c > 0$ give $\bar{k}_2 - p^c < \bar{k}_2 < \bar{k}_2 + y_2^{HS}$. This is the period-2 cost of raising the admission

probability, since a larger π_c makes the student more likely to be paying p^c instead of earning y_2^{HS}. The second term is the period-3 benefit, which is positive when $y^C > y^{HS}$. At an *arbitrary* p^c, the sign of (B24) is sign is ambiguous even when $y^C > y^{HS}$, since which term will dominate depends on the size of p^c.

By (B21), $\partial \bar{p}^c_{11|00}(p^H)/\partial \pi_c$ is $\partial \Delta / \partial \pi_c$ evaluated *at* $p^c = \bar{p}^c_{11|00}(p^H)$, which is the point of indifference (threshold) and the only point that moves when π_c changes. By (B11), $\bar{p}^c_{11|00}(p^H) < \bar{p}^c_{11|10}$ for $p^H > \bar{p}^H_{10}$, so $\bar{k}_2 - \bar{p}^c_{11|00}(p^H) > \bar{k}_2 - \bar{p}^c_{11|10}$, and by (B6) the latter equals $(\bar{k}_2 + y_2^{HS})\big((\bar{k} + y^{HS})/(\bar{k} + y^C)\big)^{\beta_3/\beta_2}$. Dividing by $(\bar{k}_2 + y_2^{HS}) > 0$, taking logarithms and multiplying by $\beta_2 > 0$,

$$\beta_2 \ln\left(\frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\bar{k}_2 + y_2^{HS}}\right) > -\beta_3 \ln\left(\frac{\bar{k} + y^C}{\bar{k} + y^{HS}}\right),$$

i.e. $\partial \Delta / \partial \pi_c > 0$ at the threshold.

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, $\partial \bar{p}^c_{11|00}(p^H)/\partial \pi_c > 0$ for every $p^H \in (\bar{p}^H_{10}, \bar{P}^H)$. If $\bar{p}^c_{11|10} \leq 0$ then $\bar{p}^c_{11|00}(p^H) \leq 0$ throughout, and the college channel is identically zero for every π_c. Hence:

$$\frac{\partial S^*}{\partial \pi_c} = \frac{1}{\bar{P}^H \bar{P}^c} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial \pi_c} \, dp^H > 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0. \tag{B25}$$

B.6.5 Effect of y^{HS}

Changes in high-school level wages operate through both the indirect labor market and the direct educational channel. Let:

$\bar{p}^H_{10} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\Pi$ with

$$\ln \Pi = \beta_2 \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k} + y^{NH}}{\bar{k} + y^{HS}}\right),$$

so $\partial \ln \Pi / \partial y^{HS} = -(\beta_2 + \beta_3)/(\bar{k} + y^{HS})$

Using $\bar{k}_1 + y_1^{NH} = 3(\bar{k} + y^{NH})$ throughout,

$$\frac{\partial \bar{p}^H_{10}}{\partial y^{HS}} = \frac{3(\bar{k} + y^{NH})(\beta_2 + \beta_3)\,\Pi}{\bar{k} + y^{HS}} \; > \; 0, \tag{B26}$$

On the direct educational channel, we have that:

$$\frac{\partial \Delta}{\partial y^{HS}} = \frac{(1-\pi_c)(\beta_2+\beta_3)}{\bar{k}+y^{HS}} \;>\; 0, \tag{B27}$$

Thus, higher high-school wages raise the high school threshold, pulling students from $(0,0)$ into $(1,0)$, and the college threshold rises too, because y^{HS} is earned in the non-admission state on the college path. Both terms of (B19) are strictly positive, so

$$\frac{\partial S^*}{\partial y^{HS}} > 0 \tag{B28}$$

B.6.6 Effect of y^{NH}

Similarly, changes in high-school dropout wages operate through both channels. Let: $\bar{p}_{10}^{H} = \bar{k}_1 - (\bar{k}_1 + y_1^{NH})\Pi$ with

$$\ln \Pi = \beta_2 \ln\left(\frac{\bar{k}_2 + y_2^{NH}}{\bar{k}_2 + y_2^{HS}}\right) + \beta_3 \ln\left(\frac{\bar{k}+y^{NH}}{\bar{k}+y^{HS}}\right),$$

so $\partial \ln \Pi / \partial y^{NH} = +(\beta_2+\beta_3)/(\bar{k}+y^{NH})$. Using $\bar{k}_1 + y_1^{NH} = 3(\bar{k}+y^{NH})$ throughout,

$$\frac{\partial \bar{p}_{10}^{H}}{\partial y^{NH}} = -3\,\Pi\,(1+\beta_2+\beta_3) \;<\; 0, \tag{B29}$$

On the direct educational channel, we have:

$$\frac{\partial \Delta}{\partial y^{NH}} = -\frac{1+\beta_2+\beta_3}{\bar{k}+y^{NH}} \;<\; 0, \tag{B30}$$

Both terms of (B19) are strictly negative, so

$$\frac{\partial S^*}{\partial y^{NH}} < 0 \tag{B31}$$

B.6.7 Effect of y^{C}

College earnings operate solely through the direct education channel. We have:

$$\frac{\partial \Delta}{\partial y^{C}} = \frac{\beta_3 \pi_c}{\bar{k}+y^{C}} \;>\; 0, \tag{B32}$$

And thus:

$$\frac{\partial \bar{p}^c_{11|00}(p^H)}{\partial y^C} = \frac{\bar{k}_2 - \bar{p}^c_{11|00}(p^H)}{\pi_c \beta_2} \cdot \frac{\beta_3 \pi_c}{\bar{k} + y^C} = \frac{\beta_3 \big(\bar{k}_2 - \bar{p}^c_{11|00}(p^H)\big)}{\beta_2 (\bar{k} + y^C)} \; > \; 0,$$

the factor π_c cancelling. Hence

$$\frac{\partial S^*}{\partial y^C} = \frac{\beta_3}{\beta_2 \, \bar{P}^H \, \bar{P}^c \, (\bar{k} + y^C)} \int_{\bar{p}^H_{10}}^{\bar{P}^H} \big(\bar{k}_2 - \bar{p}^c_{11|00}(p^H)\big) \, dp^H > 0 \quad \text{if and only if} \quad \bar{p}^c_{11|10} > 0. \tag{B33}$$

Under the necessary condition that $\bar{p}^c_{11|10} > 0$, an increase in college-level wages should increase the share of students graduating high school.